HSBC Green Finance Challenge 2025-2026

EU Climate Tech: Benchmarking the Competitive and Regulatory Environment of Climate Tech Start-Ups and Venture Capital Funds in the European Union

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1) Executive Summary

Objective

The objective of this report is to provide HSBC with comprehensive benchmarking of the Climate Tech sector in the European Union, analysing the regulatory and competitive environments in order to establish a strategic decision matrix for investment and banking support.

1. A driving but complex regulatory framework

The EU has the world's most ambitious policy framework for green transition, structured by the EU Green Deal and the "Fit for 55" package. Tools such as the European Taxonomy, SFDR, and CSRD standardize sustainable finance and create market demand for low-carbon technologies.

EU regulation is simultaneously expanding funding opportunities and increasing compliance requirements for climate tech start-ups and venture capital. Key initiatives including the Net Zero Industry Act, the Critical Raw Materials Act, the Energy Performance of Buildings Directive and the EU Venture Capital Regulation aim to strengthen clean-tech competitiveness, direct capital toward sustainable projects, and build an integrated European VC market. Overall, the framework improves transparency and capital allocation but introduces a high administrative and reporting burden on companies, creating operational complexity and compliance risks and making alignment with EU policy a decisive factor for investment and scaling.

2. The financing challenge

EU financing schemes, led by InvestEU, Innovation Fund and Horizon Europe, provide substantial public support and risk-sharing mechanisms that de-risk early innovation and expand venture activity, confirming the central role of public capital in the European climate tech ecosystem. However, the research highlights a persistent late-stage financing gap (15% of European climate techs graduated from Seed to Series B, versus 25% in the US), fragmented national participation (per-capita levels of investment remain highly unequal across EU member states) and slower capital mobilisation compared with global competitors (in 2025, EU cleantech VC investments dropped to €8.2bn versus €23.1bn in the US). These factors limit scale-up capacity despite strong growth in deal activity. For instance, VC deals in cleantech in Europe have tripled from 2015 to 2023.

Overall, public funding is a key catalyst but remains insufficient without stronger private capital mobilisation and EU-wide coordination, making financing strategy and public-private alignment decisive for investment success.

In line with these challenges, it is worth mentioning upcoming initiative in 2026 supporting the climate tech sector: the Industrial Accelerator Act, the reforms announced in the Clean Industrial Deal for 2026, the Small Modular Reactor Strategy, the Clean Energy Investment Strategy and the negotiations on the 2028-2034 EU budget will benefit start-ups based in the EU by fostering public funding, as well as expanding the demand for climate tech products.

3. EU competitive environment

The EU has major structural advantages, including an underutilized pool of innovation and a structured public support. However, the fragmentation of the single market and a financial system dominated by banks, often unsuited to companies with high intangible capital intensity, are hindering the emergence of continental champions.

The current energy transition taking place in the EU supports the Climate Tech ecosystem by providing a stronger demand, a maturity to the market and an institutional demand for climate

exposure. However the momentum is falling in 2025, with investors favoring incumbents and established champions rather than new entrants.

4. Key sectors and Strategic Case Studies

In our research, the most competitive sectors identified for investment include : Industry and energy (deep decarbonization, hydrogen, storage); Dual-use technologies (climate and defense/security); the Built environment (construction); Carbon capture (CCUS); and Critical raw materials (CRM). Our interviews with experts highlighted the fact that no sector is considered structurally unviable. However, certain sectors like hydrogen, solar and electric mobility require more selective and disciplined investment approaches.

To illustrate the reality of start-ups, the report studies the case of three start-ups: ICEYE (pioneering dual-use environmental monitoring), Vulcan Energy Resources (sustainable lithium for the battery value chain), and Stegra (large-scale green steel plan).

5. Focus on start-ups and VCs

For start-ups, the European environment provides strong structural support through regulatory stability, public grants and the rapid expansion of venture, debt and hybrid financing instruments. These factors improve visibility and early-stage de-risking. However, climate tech remains characterised by high upfront costs, capital-intensive industrialisation, persistent late-stage financing gaps, grid capacity constraints and delays linked to national regulatory implementation.

For VCs, the investment landscape benefits from structural demand for clean energy and technology alternatives, a gradual shift toward growth-stage opportunities and increasing public deployment capacity at EU level. Key risks remain global competition, supply-chain fragility and continued constraints in private capital mobilisation, which together limit the speed and scale of investment.

6. Peer benchmarking and recommendations for HSBC

Players such as BNP Paribas and Société Générale have already adopted “full life cycle” models, including direct equity investments (e.g. Solar Impulse Venture Fund) and risk-sharing partnerships with public institutions.

Finally, this report provides Strategic recommendations for HSBC. Recommendations are presented following the different themes explored throughout the report : regulation, financing schemes, competitive environment, start-ups and finally a focus is made on a case of investment.

<u>Key Strategic Recommendations</u>

- **Target the Growth Segment (Series B-C):** Align HSBC's balance sheet strength with the critical phase of industrial scaling, where the European financing gap is most pronounced.
- **Pivot to the Full Life Cycle Model:** Following the example of benchmark leaders (BNP, SocGen), move towards a hybrid offering including equity investment, structured debt, and risk sharing.
- **Systematic public-private partnerships**: Work closely with the EIB (European Investment Bank) and the EIF to raise public guarantees and reduce exposure to technological risk.
- **Adapt the risk model:** Integrate intangible assets and long commercialization cycles (~7 years) specific to climate deep tech into credit granting criteria.

In conclusion, while Europe lags behind the United States in terms of growth, regulatory stability and massive institutional support offer HSBC a unique opportunity to position itself as the preferred partner for the European industrial transition.

2) Introduction

This report aims to benchmark the Climate Technology sector in the European Union by analysing its regulatory and competitive landscape as well as current supply and demand dynamics, with the objective of building a decision matrix for investment.

It starts off by outlining the relevant regulatory and policy framework, from the EU Green Deal to key instruments such as the CSRD and the EU Taxonomy, alongside public funding and support mechanisms including InvestEU, Horizon Europe, and the Innovation Fund.

The analysis then examines the EU competitive environment and global market trends, followed by dedicated sections on start-ups and venture capital based on research and expert interviews to identify opportunities, constraints, and investment dynamics. The role of banks in supporting climate tech is subsequently assessed, with specific implications for HSBC.

The report concludes with a decision matrix summarising the key factors for strategic positioning and implementation in climate tech investments.

3) The EU Regulatory Environment and Financing Schemes

I - Regulatory and Policy Environment

A - Introduction: Why do regulations matter?

The year 2025 has been characterised by unprecedented uncertainty. A trade war launched by the US aimed at rebalancing trade flows has led to speculation about an economic slowdown. Rising inflation due to the reshaping of supply chains has resulted in turmoil on financial markets. The clean-tech sector has been heavily affected by the geopolitical situation and policy uncertainty, with a 18% decrease in real volume of investments from the first quarter of 2025 to the last quarter of 2024. More than ever, EU regulations matter for ensuring capital mobilization for VC-backed cleantech start-ups, especially considering the high-risk profile of such companies.[1]

In fact, cleantech companies face financial constraints due to the high risks and information gaps. High-capital intensity, technological complexity, elevated initial costs and the dependence on rare raw materials pose significant challenges to the scalability of cleantech start-ups. This includes the need to scale up from costly prototypes to mass production without proportional cost increases. Policy intervention is therefore necessary to make the sector more attractive to external financing, by providing credibility and transparent green-finance rules that de-risk capital.

As venture capitalists are crucial in the growth of the clean-tech sector by targeting early-stage companies and facilitating the entry of innovative companies on the market, the EU's sustainable agenda aims to deepen VC/PE markets.[2] Removing structural barriers to risk capital through the creation of a supportive framework and catalysing public investment and private investment in green innovation are key to mobilize capital in the sector.[3] Regulations and financing schemes ensure stability, directs capital to sustainable activities, and protects investors from risks that are inherent to the cleantech sector. However, an increasing number of policies has a regulatory effect

[1] Cleantech for Europe. "EU Cleantech Q1 2025 Briefing." *Cleantech for Europe*, 2025. https://www.cleantechforeurope.com/publications/cleantech-q1-briefing-2025.

[2] Ambrois, Matteo, Annalisa Croce, and Elisa Ughetto. "Greening the Future: How Venture Capital Nurtures Cleantech Companies' Growth in Europe." *Small Business Economics* 65 (January 14, 2025): 191–229. https://doi.org/10.1007/s11187-024-00988-y.

[3] Annalisa Croce et al., "Cleantech and Policy Framework in Europe: A Machine Learning Approach," *Energy Policy* 186 (March 1, 2024), https://doi.org/10.1016/j.enpol.2024.114006.

in the cleantech ecosystem limiting the number of newborn firms, while favouring the growth of existing ones.[4][5] Overregulation can stifle innovation needed for scalability, yet balanced rules unlock profitable scaling paths for VC-backed firms.

B - EU-level Regulations

At the European Union level, the main regulations concerning general environmental issues include the EU Green Deal, the Fit for 55 package, the EU Emissions Trading Scheme (ETS) and the Carbon Border Adjustment Mechanism (CBAM).

The EU Green Deal: with the aim of cutting EU emissions by 50% by 2030 (as well as other climate action-related goals), the Green Deal constitutes the overarching political and legislative background concerning renewables, energy efficiency, and Carbon Dioxide (CO2) emissions and reduction[6]. It creates the regulatory and market demand for climate technologies, while also investing in innovation, clean tech, and sustainable infrastructure (See Appendix I for further information relevant EU-level regulations.)

Beyond general policy schemes, the EU developed certain key regulations that directly concern climate tech development and investment. These make up the Sustainable Finance Framework, which aims to pinpoint sustainable economic activities and establish criteria to disclose environmentally harmful impacts.

1. The first of these regulations is the **EU taxonomy**, established in 2020 under the EU Green Deal, which aims to direct investments towards sustainable projects and activities[7]. It provides a classification system, allowing companies and other financial actors to share a common definition of environmentally sustainable economic activities. With this classification, investors and companies label their projects and outputs as environmentally sustainable (or "green"), allowing more transparency on the impacts of various products and actors.
    a. The taxonomy regulation has four overarching conditions that an economic activity must meet to qualify as environmentally sustainable:
        1) "It contributes substantially to one or more of the environmental objectives set out in the regulation
        2) It does not significantly harm any of these environmental objectives
        3) It is carried out in compliance with the minimum safeguards set out in the regulation
        4) It complies with technical screening criteria set up by the European Commission in accordance with the regulation."[8]
    b. Additionally, the regulation branches out to the **Climate Delegated Act**, which includes technical screening for economic activities to aid in a proper classification system.
    c. The EU taxonomy has direct implications for several actors, notably financial market participants offering environmentally sustainable finance or bonds; and all firms

[4] Dirk A. Zetzsche and Linn Anker-Sørensen, "Regulating Sustainable Finance in the Dark," *European Business Organization Law Review* 23 (January 24, 2022): 47–85, https://doi.org/10.1007/s40804-021-00237-9.
[5] See Footnote 2
[6] European Commission, "The European Green Deal," European Commission, 2019, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en.
[7] European Commission, "EU Taxonomy for Sustainable Activities," European Commission, 2025, https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en.
[8] Ibid.

publishing non-financial statements under the **EU Non-Financial Reporting Directive** (Directive 2013/34/EU, or Directive 2014/95/EU for the updated version)[9].

   i. The scope also extends to measures adopted by member states or by the EU that require market participants to label their financial products as environmentally sustainable.
   ii. In other words, this regulation is most relevant to financial actors and certain companies.

2. Another key regulation is the **Sustainable Finance Disclosure Regulation** (SFDR), which requires financial market participants to disclose the ways their business incorporates sustainability, and to report on sustainability risks and adverse impacts of their activities[10]. Its aim is to increase transparency in sustainability disclosures, and to help investors make informed decisions regarding the environmental costs and benefits of their funding.
3. Lastly, the **Corporate Sustainability Reporting Directive** (CSRD) provides an extension to the SFDR, requiring a much larger number of companies to report their environmental, social, and governance (ESG) impacts. Firms above a certain size are especially targeted in reporting the risks and opportunities of their business from a social and environmental perspective[11].
   a. An important principle within this regulation is that of double materiality: companies are required to report both the impact of sustainability factors on their financial performance, and the impact of their activities on the environment and society.
   b. The directive also imposes standardised disclosure, making it easier for investors to compare climate performance and assess sustainability risks and opportunities, and provide more transparent funding in the climate tech sector.
   c. As of December 2024, 15 EU member states transposed the regulation into their national legal frameworks[12].
   d. The CSRD's scope covers[13]:
      i. large companies or parent companies with over 500 employees, which were often previously subject to the NFRD;
      ii. other large companies with over 250 million employees, 50 million euros in net turnover, and/or 25 million euros in total assets;
      iii. listed small and medium-sized enterprises (SMEs), small and non-complex credit institutions, and captive insurance undertakings;
      iv. non-EU companies with significant activity in the EU.

While the SFRD and EU taxonomy focus on the financial sector, the CSRD functions at the industry level, meaning it goes beyond financial market participants to include firms in different industries.

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[9] European Parliament and Council of the European Union, "Directive 2014/95/EU of the European Parliament and of the Council of 22 October 2014 Amending Directive 2013/34/EU as regards Disclosure of Non-Financial and Diversity Information by Certain Large Undertakings and Groups Text with EEA Relevance" (2014), https://eur-lex.europa.eu/eli/dir/2014/95/oj/eng.

[10] European Commission, "Sustainable Finance Disclosures Regulation," Finance, 2023, https://finance.ec.europa.eu/regulation-and-supervision/financial-services-legislation/implementing-and-delegated-acts/sustainable-finance-disclosures-regulation_en.

[11] European Commission, "Corporate Sustainability Reporting," European Commission, 2025, https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en.

[12] Ibid.

[13] Ibid.

C - Regulations for Clean Tech Start-Ups

The aforementioned EU regulations on ESG reporting and focusing investments to environmentally conscious businesses naturally affect start-ups engaging in climate tech, both in potentially enlarging their levels of outside funding and in making the process of reporting their activities more complex.

Additionally, many EU acts and initiatives at the policy level further affect start-ups directly, and their investors indirectly.

For instance, the **Clean Industrial Deal** aims to support EU industries to decarbonise, while maintaining low energy prices and creating jobs[14]. It focuses on energy-intensive industries as well as investing in clean tech to support decarbonisation and the green transition. Relatedly, the **Green Deal Industrial Plan**, established specifically to support the development of clean tech in the EU and enhance the region's competitiveness[15]. Moreover, the **Net Zero Industry Act** (NZIA) is a framework to promote the "competitiveness of EU industry and technologies crucial for decarbonisation," and is thus directly linked to supporting the manufacturing and research of clean technologies[16].

In addition to general regulations and policies concerning environmental targets, more specialised policy schemes for specific sectors are also present. The two most relevant ones are:

- The **Critical Raw Materials Act**, concerning the sourcing and processing of critical raw materials, including lithium, cobalt, nickel, titanium, etc.[17]. Such materials are crucial for most technological products, but are often sourced and processed in precarious social and environmental contexts (often sourced in countries such as China, the DRC, or Turkiye). Hence, as of 2023, the EU has framed it as critical to improve the environmental and social footprints of its critical raw material supply chains, through recycling, different extraction procedures, and increased investment in research and innovation for critical raw material supply chains.
- The **Energy Performance of Buildings Directive**, aiming to decarbonise and decrease the energy consumption of the building sector[18].

Please refer to Appendix I for further details of relevant regulations.

D - Regulations for VCs

In the EU, several regulations and policy schemes exist to regulate VC funding and management, as well as aid them in directing their investments to relevant and sustainable projects[19]. The objective is to create an integrated European VC market. Some of the important initiatives include the three regulations under the Sustainable Finance Framework, as they play a dual role in requiring financial

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[14] European Commission, "Clean Industrial Deal," European Commission, 2024, https://commission.europa.eu/topics/competitiveness/clean-industrial-deal_en.
[15] European Commission, "The Green Deal Industrial Plan," European Commission, 2024, https://commission.europa.eu/topics/competitiveness/green-deal-industrial-plan_en.
[16] European Commission, "The Net-Zero Industry Act," Internal Market, Industry, Entrepreneurship and SMEs, 2024, https://single-market-economy.ec.europa.eu/industry/sustainability/net-zero-industry-act_en.
[17] European Commission, "European Critical Raw Materials Act," European Commission, 2024, https://commission.europa.eu/topics/competitiveness/green-deal-industrial-plan/european-critical-raw-materials-act_en.
[18] European Commission, "Energy Performance of Buildings Directive," Energy, 2023, https://energy.ec.europa.eu/topics/energy-efficiency/energy-performance-buildings/energy-performance-buildings-directive_en.
[19] OECD, "Benchmarking Government Support for Venture Capital," *OECD SME and Entrepreneurship Papers No.71* (Paris: OECD Publishing, June 27, 2025), https://doi.org/10.1787/81e53985-en.

market actors to report their impacts while also aiding investors in identifying companies' environmental footprint to better focus their investments.

Further, the **Net Zero Industry Act** (NZIA) is relevant for VCs insofar as it helps connect clean tech to investment and funding opportunities, including from private sources such as VCs[20].

Additional regulations and initiatives particularly applicable to VC funds include the **EU Venture Capital Regulation (EuVECA)** and the **Savings and Investment Union** (see Appendix I for further details).

E - Effectiveness of Regulations

The effectiveness of regulations or policies is usually measured using the various targets of each policy scheme, as well as progress on different goals and activities designed as part of the scheme.

- For the **Green Deal**, for instance, the EU set net-zero emissions as a binding target by 2050, and a 55% reduction target by 2030, and measures progress through national plans and measures adopted[21].
    - So far, current measures would lead to a net reduction of 51%, so further action is needed.
- As part of the Green Deal, the EU ETS, Energy Efficiency Directive, and Renewable Energy Directive are all binding at the EU level as well, and include Member State-level targets and commitments.
    - The **EU ETS** makes up around 45% of the EU's emissions, so is a crucial initiative to achieve the targets set out in the EU Green Deal. By 2030, the EU's aim is to achieve 62% of emission reductions through this initiative (a target changed from 43% originally, which was decided to not be ambitious enough).
    - Regarding the energy consumption under **Renewable Energy Directive**, the original goal of renewable energy comprising 42.5% of energy share in the EU does not seem likely. As few EU countries implemented renewable energy targets in line with this directive, the share of energy is on track to be between 38 and 39% by 2030, rather than the 42.5% goal[22].
- The EU taxonomy provided a useful and credible standardisation system, aiding both investors and companies in classifying green activity[23]. It also greatly improved capital allocation to green projects, and in 2025 20% of capital investments (60% of investments in utilities) aligned with the requirements set out in the regulation[24].
    - Nevertheless, it sets high bureaucratic and reporting requirements for companies which increases the complexity of operating in the EU. Its thresholds for technical

[20] See Footnote 16
[21] Liselotte Jensen and Noor Dewulf, "Enforcing EU Climate Legislation," *Briefing towards Climate Neutrality* (European Parliament, European Parliamentary Research Service, September 2024), https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762378/EPRS_BRI(2024)762378_EN.pdf.
[22] Ibid.
[23] European Commission, "Commission Provides Further Clarifications on the EU Taxonomy for Sustainable Economic Activities," Finance, November 29, 2024, https://finance.ec.europa.eu/news/commission-provides-further-clarifications-eu-taxonomy-sustainable-economic-activities-2024-11-29_en.
[24] Forest Stewardship Council International, "EU Taxonomy Regulation and Reporting Requirements," FSC, November 3, 2025, https://fsc.org/en/blog/eu-taxonomy.

performance are also incomplete and the criteria used are not strict enough to match the EU's green transition goals[25].

- Similar challenges apply to the **SFDR**, which despite increasing transparency and market discipline regarding sustainable economic activity, suffers from unclear definitions, complexity and compliance risks, and issues of "greenwashing"[26].
- Finally, the **CSRD** further increased transparency, as reports require public access and audits, as well as tangible and credible sustainability standards for investors in sustainability or climate-related projects and companies. It also helped create strategic business value regarding competitiveness of sustainability of activity, within a credible and standardised system[27]. Lastly, it contributed to increasing investments in climate and green tech through the importance of ESG goals and sustainability in the EU market, with hopes of increasing investments in R&D and deployment of climate tech as a whole[28]. The increased importance of measurement and reporting of emissions across supply chains also created new expertise and strategy focus in the market.
    - Nevertheless, it does face certain limitations, including the compliance burden imposed on companies due to the data collection and reporting process, and the complexity of implementing reporting requirements due to increasingly technical reporting, especially in relation to double materiality and value chain emissions[29]. Additionally, it is tied to regulatory uncertainty, as the EU regulatory environment surrounding climate change and sustainability reporting is constantly changing, which imposes further stress on companies.
    - Lastly, there are value chain and competition pressure from large companies and ones set up in countries with less compliance, as visualised through the **Omnibus 1** amendment to the CSRD, which limits the scope to companies with over 1000 employees or over 450 million euros in turnover[30]. This significantly decreases the scope of the regulation, and highlights the role of important companies and value chain pressures in hampering green transition efforts in the EU

Ultimately, current regulations and policy schemes in the EU face several challenges in the implementation phase and in terms of their effectiveness in reaching green transition goals. Nevertheless, they also provide an important roadmap to achieving such aims, and regulatory compliance is a crucial area for decarbonisation and emission reduction globally. As the EU allows for an extensive regulatory environment, it is important for climate tech companies and investors in the sector to align themselves with the policy and regulatory frameworks.

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[25] Franziska Schütze and Jan Stede, "The EU Sustainable Finance Taxonomy and Its Contribution to Climate Neutrality," *Journal of Sustainable Finance & Investment* 14, no. 1 (December 8, 2021): 1–33, https://doi.org/10.1080/20430795.2021.2006129.
[26] Ian Cochran, Craig Mackenzie, and Matthew Brander, "EU's Sustainable Finance Disclosure Regulation: Does the Hybrid Reporting Regime Undermine the Goal to Reorient Capital to Climate Action?," *Climate Policy* 25, no. 1 (May 22, 2024): 76–88, https://doi.org/10.1080/14693062.2024.2353115.
[27] European Innovation Council, "EIC Accelerator Open," European Innovation Council, 2023, https://eic.ec.europa.eu/eic-funding-opportunities/eic-accelerator/eic-accelerator-open_en.
[28] European Innovation Council, "EIC Accelerator Challenges 2025," European Innovation Council, 2025, https://eic.ec.europa.eu/eic-funding-opportunities/eic-accelerator/eic-accelerator-challenges-2025_en.
[29] EU Funding Portal, "EIC Accelerator 2024," EU Funding Portal, January 3, 2024, https://eufundingportal.eu/eic-accelerator-2024/.
[30] Ibid.

II - Financing Schemes

A - EU-level Financing

The EU implemented several funds or financing programmes aimed towards the green transition, many of which cover investments and funding into innovation or manufacturing of clean and climate technologies. Most of these are linked to existing regulations, such as the Green Deal, while others are EU-level initiatives dependent on funding by member states.

Key financing programmes include **InvestEU**, an initiative launched in the Green Deal in 2021 and implemented mostly by the European Investment Fund.[31] Focused on increasing investments to projects linked to sustainability, innovation, and social concerns, it guarantees support to the energy, transport, digital and circular economy sectors. It also aids in mobilising private capital and risk-sharing guarantees for projects in the climate sector.[32]

Relatedly, the **Innovation Fund**, created under the EU ETS directive and integrated into the Clean Industrial Deal, provides €40 billion from 2020-2030 to the deployment of clean and innovative technologies[33]. The funding is acquired through the EU ETS system, and is allocated through competitive tenders and auctions to climate tech in the industry, renewable energy, energy storage, carbon capture, and mobility and buildings sectors.

Another relevant financing scheme is **Horizon Europe**, a research and innovation initiative to provide support and funding (mainly in the form of grants) to various important research projects and technology development.[34] As the programme's main targets are climate change, sustainability, competitiveness, and growth; at least 30% of the budget is dedicated to climate-related projects and technologies. Adopted as part of the Green Deal and Industrial Strategy, the budget for the initiative was decided as €95.5 billion over 2021-2027, but with the newest proposal for 2028-2034 aiming to double this sum.[35]

Other financing schemes, some of which also including non-financial support, include **NextGenerationEU and REPowerEU, the European Hydrogen Bank, the Strategic Technologies for Europe Platform (STEP), the LIFE programme, and the Modernisation Fund** (Please see Appendix II for more details regarding those initiatives). Certain schemes apply to specific sectors (such as the Hydrogen Bank, focusing on hydrogen value chains), others target certain countries (such as the Modernisation Fund targeting central and eastern European countries), while many are more general towards funding and support for climate and clean technology research and development.

In addition to EU funding schemes, certain EU actors are also involved in providing funding to climate tech companies and start-ups (see Appendix II for further detail). These include:

- The **European Investment Bank**'s (EIB) climate finance targets and its climate bank roadmap, with the aim to lend for the green transition, specifically to scale climate

[31] European Investment Bank Group, "InvestEU," European Investment Bank, 2025, https://www.eib.org/en/products/mandates-partnerships/investeu/index.
[32] Ibid.
[33] European Commission, "What Is the Innovation Fund?," Climate Action, 2025, https://climate.ec.europa.eu/eu-action/eu-funding-climate-action/innovation-fund/what-innovation-fund_en.
[34] European Commission, "Horizon Europe," European Commission, 2025, https://commission.europa.eu/funding-tenders/find-funding/eu-funding-programmes/horizon-europe_en.
[35] European Commission, "Horizon Europe 2028 - 2034: Twice Bigger, Simpler, Faster and More Impactful," Research and Innovation, July 16, 2025, https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/horizon-europe-2028-2034-twice-bigger-simpler-faster-and-more-impactful-2025-07-16_en.

technologies, innovation, and infrastructure[36]. To achieve this goal, the EIB provides green financing for direct borrowers and financial intermediaries through loans, venture debt, direct equity investments, and guarantees.

- The **European Investment Fund** (EIF) provides equity investments in VC funds using public or private capital, under climate action and sustainability criteria requiring at least 90% of the VC's revenue stemming from environmentally sound activities[37].
- The **European Innovation Council** (EIC) is an initiative to support research and innovation in the EU[38]. The EIC fund provides financial support to projects and initiatives such as the STEP Scale Up and the EIC Accelerator amongst others[39]. The Accelerator provides direct equity investments into early-stage scale-ups, as well as scale-up grants and entrepreneurship advice.

B - Other Public Funds

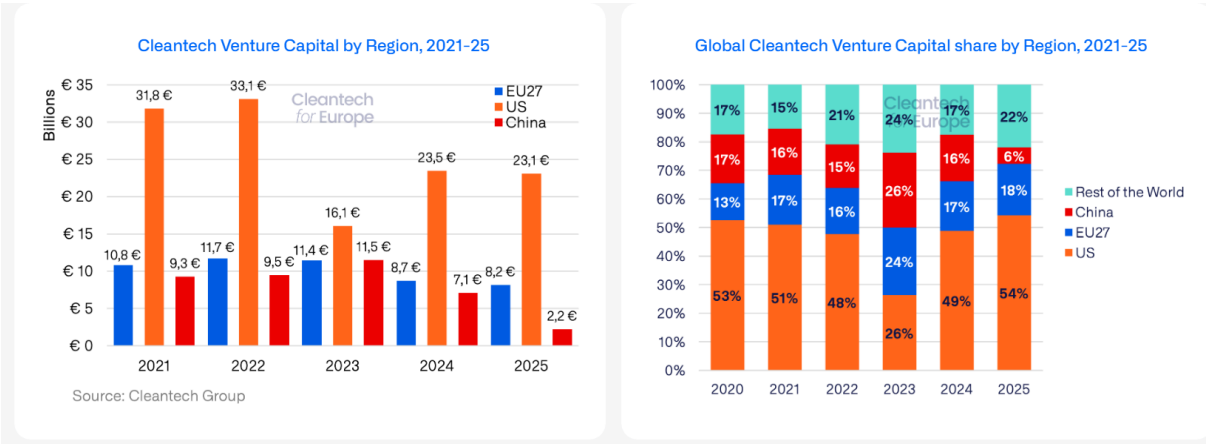
Public funds from EU Member States play a key role in supporting the expansion of clean tech companies and complement the EU's framework.

- **Bpifrance (France)** : Since 2012, Bpifrance has ranked among Europe's leading cleantech investors, alongside smart cities and impact ventures. Managing nearly €300 million, it deploys tickets from €500k to €10m in Series A–C rounds, co-investing with private capital. Today, their portfolio includes 25 start-up companies, helping them with growth acceleration and international expansion stages[40].
- **Kfw (Germany)**: In 2023, the German development bank Kfw launched the 'Green Transition Facility' program that mobilizes a 100M€ plan for green innovations. Through the creation of a new VC fund, the programme constitutes a new measure under the Future Fund and contributes to the German Government's start-up strategy,
- **LT100 Scheme (Lithuania)**: Invest Lithuania has developed financial incentives for mobilizing capital for Lithuanian-based cleantech start-ups. Through the LT100 scheme, Lithuania has allocated over €696 million to expand renewable energy and green technology development.[41]

C - Effectiveness of Financing Schemes within the European Union

a. Lagging behind other regions[42]

[36] European Investment Bank Group, "Climate and Environmental Sustainability," European Investment Bank, 2026, https://www.eib.org/en/projects/topics/climate-action/index.
[37] European Investment Fund, "Guideline on EIF's Criteria for Climate Action and Environmental Sustainability (CA&ES) Operations in EIF's Intermediated SME and Mid-Cap Financing" (European Investment Fund, April 2022), https://www.eif.org/files/public-register/climate-action-sustainability-criteria.pdf.
[38] European Innovation Council, "European Innovation Council," European Innovation Council, 2026, https://eic.ec.europa.eu/index_en.
[39] European Innovation Council Fund, "About the EIC Fund," European Innovation Council, 2026, https://eic.ec.europa.eu/eic-fund/about-eic-fund_en.
[40] Bpi France, "Cleantech, Smart City & Impact Funds," Bpi france, March 7, 2022, https://www.bpifrance.com/products/cleantech-smart-city-impact-funds/.
[41] Invest Lithuania, "Cleantech in Lithuania," Invest Lithuania, February 5, 2026, https://investlithuania.com/cleantech/.
[42] Cleantech for Europe, "EU Cleantech Annual Briefing," *Cleantech for Europe*, 2025, https://cleantechforeurope.com/publications/cleantech-annual-briefing-2025.



In 2025, EU cleantech VC investments continued to lag global leaders, dropping to €8.2bn versus €23.1bn in the US, while China posted a second straight year of steep decline to €2.2bn. In 2025, the EU's share of global cleantech venture and growth investment rose slightly to 18%, yet remained small compared to the US dominance reflected by a 54% share. One difference is Governments, especially in Europe and Canada, have taken an increasingly active role in VC markets, whereas the VC industries of the United States are largely driven by the private sector.[43] In fact, Europe's VC ecosystem faces a huge public-private financing gap. European startups that are in their early business phases have relatively good access to seed capital. However, when they enter the scaleup phases, many European startups struggle to raise the capital they need to expand their production.

b. The weaknesses of the EU zone

Europe has been criticized for being "too slow" in mobilizing capital for cleantech companies. The CleanTech for Europe's 2025 Outlook points out 2025 as the second consecutive year of declining venture and growth equity, and persistent late-stage financing gaps proving the lack of manufacturing industries in Europe necessary to scalability.[44]

(i) The private sector funding gap

The EU is facing a worsening Series B funding gap compared to competitors[45]. From 2020 to 2024, Europe invested $34 billion in early-stage climatetech firms, yet many faltered at Series B. The situation is exacerbated by the fact that more startups are now reaching this stage without adequate follow-on funding. In fact, 15% of European climate techs graduated from Seed to Series B, vs 25% in the US. European venture capital investors remain cautious about climatetech due to its capital-intensive profile. Unlike software ventures, which scale with minimal marginal costs and offer rapid exits (typically 3–5 years), climatetech requires substantial upfront spending on prototypes, facilities, and supply chains.

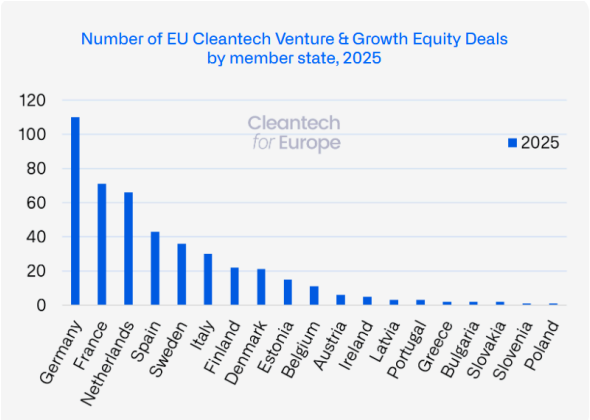
(ii) Industrial policy shortfalls

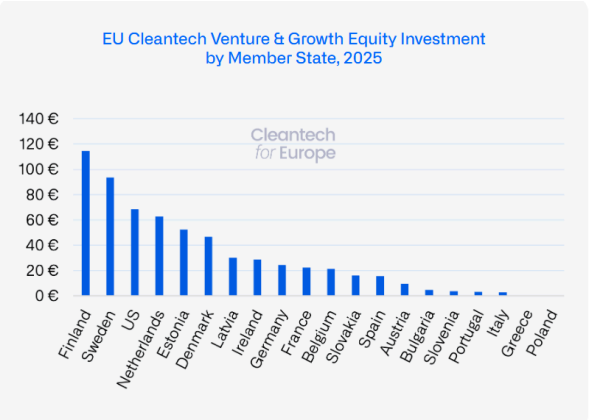
Numerous partners including India, Brazil, and the United States are actively implementing industrial policy instruments, such as local content requirements, to develop domestic clean technology industries (ie. US IRA).[46] Europe's public funding mechanisms lag behind competitors. National

[43] See Footnote 19
[44] See Footnote 42
[45] Veronica Fresneau, "The Series B Funding Gap in European Climate Tech," World Fund, 2020, https://www.worldfund.vc/articles/the-series-b-funding-gap-in-european-climate-tech.
[46] Max Gruenig, "The US Inflation Reduction Act: Impacts on Cleantech, Trade and Investment," E3G, September 23, 2025, https://www.e3g.org/publications/the-us-inflation-reduction-act-impacts-on-cleantech-trade-and-investment/.

sovereign funds are modest compared to the US's ARPA-E or China's state-directed industrial complexes. Strengthening initiatives like France 2030 across the EU could address these structural deficits[47].

(iii) Lack of coordination among MS[48]





Cleantech VC deals took place in 19 of 27 EU member states in 2025. Germany led with 110 deals, followed by France (71), the Netherlands (66), Spain (43), and Sweden (36). Per-capita levels of investment remain highly unequal across EU member states with Finland (€115) and Sweden (€94) recording the highest cleantech investments per capita. Most countries remained well below these levels: the Netherlands (3rd place) captured only about half of Finland's per capita cleantech investments.

Moreover, the existence of national public funds (see part B) among EU Member States suggests that :

1) Beyond the EU's framework, national strategies are effective in helping the expansion of early-stage clean tech startups.
2) However, the financing gap between Member States suggests a higher coordination effort and broadening participation remains a key challenge for the EU's ambitions. Uneven participation and regulations across EU countries mitigate the EU's cleantech strategy.[49]

c. EU's cleantech environment: a positive outlook amid challenges

VC deals in clean- tech in Europe **have tripled from 2015 to 2023**, peaking in 2021 and 2022. The sector expanded rapidly in 2023, with 60% more deals being closed than in 2019. This growth was supported by the development of a framework that was favourable to the EU focusing on de-risking and scalability. [50]

- For example, the **EU Innovation Fund** shows a good track record with 77 decarbonisation projects and grant agreements in 2025. The initiative has shown potential to reduce emissions by 397 million tonnes.[51] In 2024, 36 projects were fully

[47] Gouvernement France, "France 2030 : Un Plan d'Investissement Pour La France," economie.gouv.fr, October 13, 2023, https://www.economie.gouv.fr/france-2030.
[48] See Footnote 1
[49] See Footnote 1
[50] See Footnote 2
[51] European Climate, Infrastructure and Environment Executive Agency, "Innovation Fund: €4.2 Billion to Support 77 Cutting-Edge Decarbonisation Projects for EU's Clean Energy Transition," European Commission, March 11, 2025, https://cinea.ec.europa.eu/news-events/news/innovation-fund-eu42-billion-support-77-cutting-edge-decarbonisation-projects-eus-clean-energy-2025-03-11_en.

funded and 7 were successfully deployed.[52] However,the project has shown issues with the timeline, as projects need to reach "financial close" within 4 years from grant signature.[53] Securing off-take agreements, especially in nascent markets with market uncertainties and limited competitiveness inherent to innovative green-products compared to conventional alternatives, has posed significant challenges. With inflation rising and the reshaping of world supply chains, issues of cost have become prominent combined with supply chain constraints. Institutional fragmentation, procedural lag, and intra-EU competition continue to mitigate impact and undermine strategic coherence. Competition issues, with competition both within the market and from existing market players/technologies.[54]

- The **EIF** has also contributed to the expansion of the EU VC market.[55] By assuming the risk that the fund will bear when starting operations with less capital, it helps diffuse international best practices and provide advisory services to VC funds willing to invest in cleantech within the European Union.

III - The Future of Policy

In line with the current challenges facing regulatory and policy initiatives in the EU climate tech sector, 2026 will see several new schemes concerning the sector being adopted. Certain upcoming initiatives relevant to start-ups and VC funds in clean tech include:

- The **Industrial Accelerator Act**, planned to be adopted on 25 February 2026, would extend the Clean Industrial Deal by facilitating faster and easier permitting for decarbonisation of industries, and establishing a low-carbon level for certain sectors such as steel and cement[56]. The aim is to facilitate the protection of EU clean industrial products.
- Reforms of the **Clean Industrial Deal** in 2026 are on track to add an Industrial Decarbonisation Bank, based on the funds in the Innovation Fund with revenues from the ETS and InvestEU programmes[57]. It will also include the Circular Economy Act, which aims to stimulate demand and supply of circular (sustainable) products with the target of 24% of EU materials coming from circular sources by 2030[58]. Lastly, the policy scheme will create a

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[52] European Commission and European Climate, Infrastructure and Environment Executive Agency, "2025 Annual Knowledge Sharing Report of the Innovation Fund," *European Commission* (Publications Office of the European Union, July 2025), https://op.europa.eu/webpub/clima/innovation-fund-2025/en/.
[53] Ibid.
[54] European Commission, and Directorate-General for Climate Action. "Innovation Fund 2024: Investing into the Future of Net-Zero Technologies and Electric Vehicle Battery Cell Manufacturing." Climate Action, May 14, 2025. https://climate.ec.europa.eu/news-other-reads/news/innovation-fund-2024-investing-future-net-zero-technologies-and-electric-vehicle-battery-cell-2025-05-14_en.
[55] Sebastian Mack, "Europe Ventures Forward: Getting the Scaleup of Cleantech Right," *Hertie School Jacques Delors Centre* (Hertie School, October 30, 2024), https://www.delorscentre.eu/en/publications/detail/publication/venture-capital-getting-the-scaleup-of-cleantech-right.
[56] European Parliament, "Industrial Accelerator Act," European Parliament Legislative Train Schedule, January 22, 2026, https://www.europarl.europa.eu/legislative-train/theme-a-new-plan-for-europe-s-sustainable-prosperity-and-competitiveness/file-industrial-decarbonisation-accelerator-act.
[57] European Innovation Council, "EIC 2025 Work Programme," European Innovation Council, October 29, 2024, https://eic.ec.europa.eu/eic-2025-work-programme_en.
[58] See Footnote 52

Critical Raw Materials Centre, to purchase raw materials jointly and create economies of scale to facilitate the purchase prices and conditions of such materials[59].

- The **Small Modular Reactor** (SMR) **Strategy**, to be published in the first half of 2026, addresses the EU's aims to be carbon-neutral by 2050, to improve economic competitiveness, and to phase out Russian energy imports[60].
- The **Clean Energy Investment Strategy**, Grids Package, Citizens Energy Package and Energy Omnibus (which will simplify existing legislation) are all initiatives to be published in 2026 focusing on the energy sector[61].
- Negotiations on the **2028–2034 EU budget** will intensify along 2026. The outcome will have a significant impact on European cleantech financing strategy. As public finances are being contracted, the budget allocated to green innovations will depend on the European Union's top priorities[62].

For climate tech, the upcoming EU initiatives offer the opportunity for further public funding and support, as well as more demand for sustainable or circular products (and the green technologies associated). Furthermore, the new policies aim to benefit start-ups based in the EU or with diversified supply chains, thus improving the competitive advantages of EU climate tech.

Other reforms to energy directives, the ETS and CBAM regulations, and other environmental policies will further expand the market for climate tech to help with the EU's green transition and growing market for renewable energy, clean transport and building, and other green products[63].

4) The EU Competitive Environment

This section aims to answer the question: what is the competitive environment for climate tech VCs in Europe? First, we describe the structural elements of the EU competitive environment and its advantages and weaknesses (I), in the second part we develop the opportunities and threats linked to the actual investment trends (II). Please refer to Appendix III for charts relevant to this section.

I - Structural elements

A - Competitive Advantages

1) Policy and Regulatory Support

[59] Mahnoor Syed, "European Innovation Council Fuels Deep Tech Startups for 2025, Singapore Launches Climate Ventures to Propel Southeast Asia's Climate Tech Innovation & More," Climate Insider, October 30, 2024, https://climateinsider.com/2024/10/30/european-innovation-council-fuels-deep-tech-startups-for-2025-singapore-launches-climate-ventures-to-propel-southeast-asias-climate-tech-innovation-more/.

[60] European Commission and Directorate-General for Energy, "Commission Invites Views to Shape Its Small Modular Reactor Strategy," Energy, November 10, 2025, https://energy.ec.europa.eu/news/commission-invites-views-shape-its-small-modular-reactor-strategy-2025-11-10_en.

[61] See Footnote 56

[62] See Footnote 42

[63] European Commission, "Innovation Fund 2025 Net-Zero Technologies Call," European Climate, Infrastructure and Environment Executive Agency, December 4, 2025, https://cinea.ec.europa.eu/funding-opportunities/calls-proposals/innovation-fund-2025-net-zero-technologies-call_en.

The EU has a strong political and social commitment to clean energy and net-zero targets, which is reflected in its policy and regulatory landscape[64]. Broad climate initiatives such as the EU Green Deal, EU Emissions Trading Scheme (ETS) or the Clean Industrial Deal all support a shift towards the adoption of increasingly low-carbon technologies, and provide backing for the climate tech sector[65]. At the same time, regulation should remain simple and introduce inflexible permitting procedures or else risk becoming barriers to clean tech deployment[66].

2) Structured public support

In Europe, powerful public institutions such as the EIF, EIB, and national development banks are present. For private investors, they reduce risk in the early stages, act as anchor LPs and encourage the emergence of local funds[67]. This stabilises venture capital at the bottom of the cycle, a relative advantage over purely private markets. For example, the EIB was the most active venture capital investor in climate tech in 2025.

3) Less competition in late-stage

Europe is characterized by a low density of domestic mega-funds, reducing competitive bidding pressure on certain assets[68]. VCs able to mobilize international capital therefore hold a strategic positioning advantage.

4) Underutilised reservoir of innovation

Europe produces a great deal of research, technical talent and deep tech start-ups (climate, industrial, health). For VCs, this creates a dense deal flow at lower valuations than in the United States.

B - Structural Weaknesses

1) Fragmentation of the single market and lack of proper capital markets union

The first issue for Europe is the fragmentation of its capital markets. The EU is composed of 27 member states, only 20 of which share a currency, speaking 24 languages. Financial supervision, regulatory harmonisation, taxation, labour law, capital markets and languages remain largely national, impacting the level of economic integration between EU members[69].

For VCs, this creates:

- high scaling costs (to adapt to each national framework)
- more complex pan-European theses
- difficulty in building continental champions

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[64] McKinsey, "Bold Moves, Fast Scale-Up: Europe's Path to Cleantech Competitiveness," *McKinsey Sustainability* (McKinsey & Company, November 25, 2025), https://www.mckinsey.com/capabilities/sustainability/our-insights/bold-moves-fast-scaleup-europes-path-to-cleantech-competitiveness.

[65] European Commission and Directorate-General for Energy, "2025 Competitiveness Progress Report: Opportunities to Boost EU Clean Tech," Energy, February 26, 2025, https://energy.ec.europa.eu/news/2025-competitiveness-progress-report-opportunities-boost-eu-clean-tech-2025-02-26_en.

[66] See Footnote 64

[67] Nathaniel Arnold et al., "Stepping up Venture Capital to Finance Innovation in Europe: WP/24/146," *International Monetary Fund* (International Monetary Fund, July 2024), https://www.imf.org/-/media/files/publications/wp/2024/english/wpiea2024146-print-pdf.pdf.

[68] Sightline Climate, "The Globalization of Climate Tech," *HSBC* (HSBC Innovation Banking, November 11, 2025), https://www.hsbcinnovationbanking.com/en/resources/the-globalization-of-climate-tech.

[69] See Footnote 67

This has triggered the debate over a 28th regime, showing also that the issue has been identified and the situation for start-ups could be improved.[70]

2) Financial system dominated by banks

The second biggest issue to explain the weaknesses of the EU environment for VCs is the structure of the financial system (cf annex figure 5). According to the IMF, banks are ill-suited to financing startups for at least four reasons[71]:

- The important role of collateral in bank risk management works against new firms with predominantly intangible assets (R&D, patents, and intellectual property more broadly) because such assets are not easily attached, seized, or sold.
- Many bank risk models are not well attuned to the knowledge intensive and subjective task of gauging the future prospects of high-tech fast-growing but initially unprofitable firms.
- The time it takes startups to develop products, scale up, and generate sufficient revenue is generally not consistent with banks' maturity and debt-servicing requirements.
- Regulation and supervision require high-risk exposures to be amply buffered by capital and provisions, which can significantly reduce banks' rates of return on loans to startups.

It results in an underfunding of equity, excessive dependence on subsidies and public debt, and less depth of funding rounds.

On the start-up side, the leading role of banks in funding climate tech diminishes the potential importance of public support, including tax breaks or subsidies for certain technologies or infrastructure[72]. This decreases EU competitiveness in the field compared to other global actors, such as China or the US, where public support plays a large role.

Several experts shared with us the unadaptability of banks to start-ups like tech companies. These types of companies are smaller than the size that banks are generally comfortable with. Models are not adapted to these companies which have to wait around 2-3 years to show some returns. **Banks' internal risk models may not be adequate** for investments in such companies. Which links to another issue, **investors in the EU remain too risk averse.** While regulation, such as strict capital requirements, does play a role, it seems that the real bottleneck is not due to regulation but rather **risk aversion.**

3) Small pools of private capital

European savings stay mainly in bank deposits, with pension funds and insurers having little exposure to venture capital. In Europe, assets held in private pension funds and insurance companies collectively amount to some $11.9 trillion, whereas it is roughly $42.5 trillion in the US (cf annexes Figure 15)[73].

The results for venture capital in Europe:

- smaller funds
- difficulty in raising late-stage vehicles
- persistent scale-up financing gap

[70] Hallak Issam, "The 28th Regime: A New Legal Framework for Innovative Companies," European Parliament Think Tank, January 12, 2026, https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2026)782601.
[71] See Footnote 67
[72] See Footnote 64
[73] See Footnote 68

On this topic, European savings are estimated at €35.5 trillion but are insufficiently mobilised. This led the European Commission to publish in March 2025 its recommendations for a strategy for a Savings and Investment Union[74].

4) Unattractive exit markets
EU IPOs are fragmented, less liquid, with lower valorisations compared to the US[75].
For VCs, it implies lower expected returns, and an incentive for start-ups to seek funding abroad and sometimes relocate outside Europe.

5) The loss of London as a European VC hub
Brexit resulted in the loss of Europe's only VC hub with global-scale significance, leaving remaining centers such as Paris, Berlin, Amsterdam, and Stockholm without sufficient critical mass.

II - Advantages and Weaknesses Regarding the Trends

A - Actual trends

In addition to sightline reports[76]:
(general, not only Climate tech) In Europe, figures for H1 2025 confirm a sharp contraction in venture capital: 2,383 deals recorded (-44% vs H1 2024) for €24.6 billion raised (-12%), but with an average ticket soaring to €10.32 million (+57%). This trend reflects a profound shift in investment: funds are refocusing on more mature growth operations, in Series B or higher, in sectors considered strategic such as artificial intelligence, cybersecurity and climate infrastructure.

This concentration of investment in a few highly capitalised players reflects a dual requirement: project maturity and alignment with major geopolitical or climate issues.

Investment in cleantech and energy declined at the European level in H1 2025: -52% in amounts raised and -53% in number of deals compared to H1 2024. Investors are refocusing on more mature industrial projects (Series B/C), concentrated on energy infrastructure, industrial decarbonisation and hydrogen[77].

B - Opportunities

1) Structural demand shift: energy, resilience, and security backed by a political framework aligned on structural transitions
EU priorities like climate action, as well as energy, industry, and technological sovereignty, create:

- a regulatory demand for certain technologies,
- long-term visibility on investment theses (clean tech, industrial tech).

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[74] European Commission and Directorate-General for Communication, "Savings and Investments Union: Better Financial Opportunities for EU Citizens and Businesses," European Commission, March 19, 2025, https://commission.europa.eu/news-and-media/news/savings-and-investments-union-better-financial-opportunities-eu-citizens-and-businesses-2025-03-19_en.
[75] John MacDonagh and Oscar Allaway, "Emerging Tech Research: VC Investment in Climate Tech," *PitchBook* (PitchBook Data Inc, June 24, 2025), https://files.pitchbook.com/website/files/pdf/Q2_2025_PitchBook_Analyst_Note_VC_Investment_in_Climate_Tech_19518.pdf.
[76] In Extenso, "[Revue de Presse] Baromètre Des Levées de Fonds – S1 2025 - in Extenso Innovation Croissance," In Extenso Innovation Croissance, October 2, 2025, https://www.inextenso-innovation.fr/revue-de-presse-barometre-des-levees-de-fonds-s1-2025/.
[77] Ibid.

It's an advantage for specialised VCs. Climate tech investment has moved away from abstract decarbonisation targets toward hard constraints (power availability, grid stability, energy security, and climate adaptation)[78].

Increased public demand for net-zero technology is also an advantage for start-ups, giving them access to a larger market with increased demand for various types of climate tech[79].

This context favors European strengths :

- rising electricity demand (AI, data centers, electrification) creates non-cyclical demand for firm power, grid flexibility, storage, and backup solutions.
- Energy security concerns (Ukraine war, gas dependence, critical minerals) align strongly with European policy priorities.
- Technologies are now valued for speed-to-power, reliability, and cost, with emissions reduction increasingly treated as a secondary (but still positive) attribute

During our interviews, the sifting moment for Europe was stressed : investors start to see the EU as an attractive market. From a US perspective, Europe appears more resilient and less volatile, in the sense that the EU's policy direction remains relatively stable, contrary to the US.

2) Maturity of the market and rise of growth-stage capital

The market is consolidating around fewer, larger, more credible bets. Deal count is falling, but total capital is rising. Series B increasingly resembles late-stage investment in size and intent. Investors are prioritizing companies with proven deployment paths, regulatory traction, and industrial partnerships[80].

For Europe, this creates opportunity: the ecosystem produces technically credible companies, often spin-outs from industry or research.

This environment rewards capital providers with balance-sheet depth, risk-structuring capability, and growth financing expertise (generally positive for banks).

3) Institutional and LP-driven demand for climate exposure

Beyond pure financial returns[81]:

- LPs are using climate specialist funds to align portfolios with internal ESG and decarbonisation objectives
- Even amid regional VC weakness, global interest in climate resilience and energy security remains strong

This supports continued relevance of European climate VC as a strategic asset class and a demand for structured products, co-investments, fund finance, and growth capital solutions rather than pure early-stage risk.

C - Weaknesses and Competitive Disadvantages of Europe

1) Persistent capital decline and loss of momentum

Europe continues to underperform materially[82]:

[78] See Footnote 75
[79] See Footnote 65
[80] See Footnote 67
[81] Ibid.
[82] See Footnote 75

- VC climate investment fell 13% YoY in 2025 to $10.1bn, the lowest since 2020
- While global climate tech investment remains historically high, Europe is losing share relative to the US and select Asian markets
- Fundraising momentum is weakening further in early 2025, partly due to broader VC cycles but amplified by regional fragility

This reinforces a **structural capital gap**[83]:

- Particularly for Scale-up and late-stage growth rounds
- For capital-intensive technologies requiring long deployment timelines.

Moreover, supply chain dependencies and weak value chains, linked to high production costs and labour shortages, contribute to the EU's loss of momentum in the production and deployment of clean tech[84]. Those factors in addition to increased global competition in the sector decreases the perceived value of investing in innovation or production of climate tech in the EU.

2) Market consolidation favors incumbents, not new entrants

Risk appetite has compressed:

- "Champions" are already chosen in sectors such as hydrogen, SAF, and carbon removal
- Investors are concentrating capital into companies with established track records
- Fewer first-time founders and new technical entrants are being backed

For Europe, this is particularly damaging[85]:

- The ecosystem already struggles to scale new entrants into global leaders
- Capital concentration reinforces path dependency and limits renewal

3) Macro and thematic risk exposure

Two major risks dominate forward-looking assessments:

- **AI-driven energy demand** may partially reflect speculative excess, thus a correction could hit power companies in the climate tech sector
- Climate VC remains exposed to policy volatility, especially as US federal support becomes less predictable and global trade tensions rise

Europe's reliance on public policy alignment also limits flexibility and speed compared to more market-driven ecosystems.

But these threats are also working for a European momentum: Project development is shifting to Europe and emerging markets. The promise of faster permitting, cheaper feedstocks, and massive scale is raising interest in deploying climate tech outside of the US. The European Investment Bank has €1tn ($1.2tr) to spend before 2030, giving a strong signal for increased climate tech investment outside the US[86].

[83] European Commission and Joint Research Centre, "Climate-Neutral Tech Solutions: EU's Strong in Innovation, but Lags in Commercialisation - European Commission," Joint Research Centre, November 15, 2023, https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/climate-neutral-tech-solutions-eus-strong-innovation-lags-commercialisation-2023-11-15_en.

[84] See Footnote 65

[85] See Footnote 83

[86] See Footnote 67

5) Start-Ups: Sectorial Overview

I - Relevant and Competitive Sectors

Starting with the sectorial divisions from the Sightline report (see Appendix IV for detailed information), and following further research, the following list of most relevant climate tech sectors at the EU has been compiled[87], based on the current importance, profitability, and competitiveness of those sectors in the EU compared to elsewhere[88].

1) **Industry** is considered one of the most competitive sectors at the EU level in the context of climate tech, attributed to the efforts of EU regulations in decarbonising and innovating the sector to reach net-zero emissions[89].
    a) Several EU regulations are aimed towards the industrial sector, including the Clean Industrial Deal, the Innovation Fund, and the Net Zero Industry Act. Those policies all set targets for the decarbonisation of the sector and its transition towards cleaner production, which is at the core of the EU's green transition plan. As industry is considered one of the main targets for decreasing emissions, climate tech geared towards industry is a crucial and currently competitive sector in the EU.
    b) Nevertheless, the sector faces certain challenges with high capital costs of decarbonising very energy intensive industries, as well as high global competition with countries such as China or the U.S., who face less complex and constraining environmental regulations.
2) **Energy** is a competitive sector at the EU level insofar as there has been a large amount of funding in various renewable energy companies. Renewable energy has consistently been one of the EU's primary targets for the green transitions, and has thus received a lot of investment, research, and policy support.
    a) Regulations offering support to the energy sector include the Modernisation Fund, Innovation Fund, the European Hydrogen Bank, the Fit for 55 package of the EU Green Deal, etc.
    b) One of the main limitations of the sector is high competition from China (in terms of production and deployment costs) and the U.S. (in terms of research and technological advances). Furthermore, the EU requires increased investment in the sector for research and development, and suffers from commercialisation and funding issues[90].
    c) It is also important to note that energy is one of the sectors suffering the most from inertia and path-dependency, which may hamper the successful development of climate technologies[91].
3) **Dual-use tech**: Dual-use technology refers to start-ups that can employ their product in civilian as well as military space. Amid the geopolitical uncertainties, Europe's rearmament reinforces the investment case for dual-use climate-tech startups. In areas such as resilient clean power, energy storage and portable renewables we find that the same technologies can

[87] See Footnote 64
[88] Bruegel, "European Clean Tech Tracker," Bruegel Clean Tech Tracker, 2025, https://european-clean-tech-tracker.bruegel.org/.
[89] European Commission, "Industry," Internal Market, Industry, Entrepreneurship and SMEs, 2026, https://single-market-economy.ec.europa.eu/industry_en.
[90] See Footnote 65
[91] Jan Christoph, "The State of European Climate Tech in 2025: Is Climate Tech Dead?," EU-Startups, August 8, 2025, https://www.eu-startups.com/2025/08/the-state-of-european-climate-tech-in-2025-is-climate-tech-dead/.

serve both defence needs and the green transition, creating scalable demand across public and private markets.[92]

a) There are growing signs that this is a promising sector for investment. For instance, 2025 has seen the investments in dual-use and defence technology reach $3.88 billion.[93] There is initiative also on the EU level with the European Investment Fund wanting to mobilise about 500 million in private capital for dual-use startups.[94]

b) While not all of it includes climate technology, there is overlap in several domains. One important area is space-based systems, where climate and defence needs strongly converge. Earth-observation satellites and data-processing platforms are essential for environmental monitoring, climate-risk assessment, and disaster response, while simultaneously underpinning military situational awareness and operational resilience.[95] Satellite startups such as ICEYE or drone-focused Delian Alliance Industries are just a couple of successful examples.[96] At the same time, decentralised clean-energy technologies, including energy storage and portable renewables, enhance civilian energy security while increasing the resilience and autonomy of military infrastructure. The relevance of this overlap is visible at the national level. For example, the Spanish government has designated €1.75 bln of its military budget for climate-responsive dual-use capabilities.[97] This was due to how climate change affects militaries. For instance, in 2024 major floods tied up thousands of troops in Spain, which held back the military's readiness for core defence tasks.[98]

4) The **Built Environment**, consisting of any construction for human use and lifestyles, is one of the most regulated sectors in the EU due to its high emissions and other environmental impacts[99]. Moreover, climate change has also been proven to affect the built environment and construction sectors, especially in relation to energy usage, making it an especially sensitive area in the context of climate change[100]. Since the built environment is crucial for everyday life, the EU has put a special focus on increasing the sustainability of the sector[101]. The ever-changing regulations, initiatives, and discussions about making this sector more green

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[92] Simone Tagliapietra, "Defence and Climate: Seven Points for a Common Agenda," *Bruegel* (Bruegel, April 28, 2025), https://www.bruegel.org/analysis/defence-and-climate-seven-points-common-agenda.
[93] Lisa Palchynska, "How to Scale Europe's Dual-Use and Defence Tech: Insights from the Founder's Handbook 2.0," Vestbee, September 30, 2025, https://www.vestbee.com/insights/articles/dual-use-founder-s-handbook.
[94] Ibid.
[95] ICEYE, "ICEYE Recognized in the World's 50 Most Innovative Companies List by Fast Company," ICEYE, March 18, 2025, https://www.iceye.com/newsroom/press-releases/iceye-recognized-as-one-of-the-worlds-50-most-innovative-companies-by-fast-company.
[96] Teodora Atanasova, "Top 6 Defense and Dual-Use Tech Startups from CEE," The Recursive, July 17, 2025, https://therecursive.com/top-defense-dual-use-startups-cee/.
[97] Florian Krampe, "Europe's Defence Push Can't Ignore Dual-Use Climate-Defence Tech," World Economic Forum, July 31, 2025, https://www.weforum.org/stories/2025/07/europe-defence-build-climate-dual-use/.
[98] Ibid.
[99] Bawantha Rathnayaka et al., "Climate Change Impacts on Built Environment: A Systematic Review," in *Lecture Notes in Civil Engineering*, 2022, 443–59, https://doi.org/10.1007/978-981-19-2886-4_31.
[100] R.L. Wilby, "A Review of Climate Change Impacts on the Built Environment," *Built Environment* 33, no. 1 (March 13, 2007): 31–45, https://doi.org/10.2148/benv.33.1.31.
[101] McKinsey, "Igniting Opportunities in the EU's Built Environment," McKinsey & Company, February 28, 2023, https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/refurbishing-europe-igniting-opportunities-in-the-built-environment.

while still making it livable and efficient makes it one of the most crucial and evolving sectors for climate tech and green investments at the EU level[102].

a) Regulations offering support to the construction and built environment sector include the Innovation Fund, Energy Performance of Buildings Directive, and the Fit for 55 package of the EU Green Deal.
b) There is currently an important investment gap and inertia in the sector due to the difficulty of implementing standards and regulations while maintaining activity necessary for everyday life. There are also high upfront costs in decarbonising the sector, and the market is highly fragmented.

5) **Carbon** capture, usage, and storage is a relatively novel sector, but has received considerable attention especially within the EU. As it is a rapidly involving sector, requiring a lot of research and innovation, it is both highly competitive and much riskier than other sectors due to a higher degree of uncertainty in the technologies developed and their efficiency[103]. Nevertheless, a rise in funding for research and deployment of carbon-related technologies makes it a crucial climate tech sector (especially due to the effectiveness of developed technologies in aiding climate mitigation)[104]. Through several regulations, including the ETS, and initiatives to develop carbon capture and storage technologies, the EU is currently one of the world leaders in carbon technologies[105].
    a) Regulations geared towards the carbon sector include the Innovation Fund and the Net Zero Industry Act. The sector has received less policy and regulatory attention than other areas, but remains a crucial part of the EU green transition strategy.
    b) The carbon sector faces severe challenges in profitability and funding, as investments are seen as risky with often no guarantee of the technologies being effective and profitable. High return to investment is often needed for investors, making public funding and blended finance essential for current projects.
6) **Critical Raw Materials** (CRM), since 2023, has been increasingly seen as an important sector for EU supply chain management, including in climate neutrality targets[106]. Such materials are crucial both economically and for technology development worldwide, and their extraction and distribution is highly concentrated, which constitutes a high risk of supply chain shocks. The EU thus aims to diversify its CRM supply while simultaneously improving the environmental and social costs of the sector. The 2024 Critical Raw Materials Act was passed as a support for those aims and to funnel investment into relevant research, technology, and projects in the sector.

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[102] European Commission, "What Is BUILD UP?," BUILD UP, 2026, https://build-up.ec.europa.eu/en/about/whats-is-bup.
[103] Jing Meng et al., "Technologies and Gaps in Deep Decarbonization of Hard-To-Abate Industrial Sectors," *Nature Reviews Clean Technology* 1 (June 30, 2025): 578–95, https://doi.org/10.1038/s44359-025-00082-w.; Zihe Wang et al., "An In-Depth Review of Key Technologies and Pathways to Carbon Neutrality: Classification and Assessment of Decarbonization Technologies," *Carbon Neutrality* 4, no. 1 (June 4, 2025), https://doi.org/10.1007/s43979-025-00129-8.
[104] Alicia García-Herrero, "Different Routes to Decarbonisation: EU Carbon Pricing versus China's Green Tech," Bruegel, November 10, 2025, https://www.bruegel.org/newsletter/different-routes-decarbonisation-eu-carbon-pricing-versus-chinas-green-tech.
[105] European Commission, "Trends in Carbon Intensity and the Macroeconomic Role of the EU Emissions Trading System," Economy and Finance, 2025, https://economy-finance.ec.europa.eu/trends-carbon-intensity-and-macroeconomic-role-eu-emissions-trading-system_en.
[106] European Council, "An EU Critical Raw Materials Act for the Future of EU Supply Chains," European Council, March 21, 2025, https://www.consilium.europa.eu/en/infographics/critical-raw-materials/.

a) There is a lot of investment, both public and private, into research, innovation, and strategic projects in the sector[107]. Thus, it is currently a rapidly growing and interesting sector for investments.
b) However, CRM investments are also considered highly risky and uncertain, so has not seen as much project development in terms of climate tech as other, better established sectors[108].

Key takeaway from our expert interviews: no sector is viewed as irreversibly "lost" within the European climate tech ecosystem. While certain segments are regarded as particularly promising - notably energy storage and batteries (capital-intensive yet essential) and carbon dioxide removal (CDR), driven by the expansion of carbon credit markets - others warrant a more selective and disciplined investment approach, including hydrogen, solar, and electric vehicles (EVs).

II - Relevant Actors / Case studies

1) ICEYE

Named by Times and Statista as among the top 250 greentech companies of 2025, this Finnish start-up is using satellite imagery for various purposes from environmental monitoring to military reconnaissance.[109] ICEYE has established itself as a key environmental monitoring provider by applying SAR satellite technology to climate-related risk assessment and disaster response, enabling continuous observation regardless of weather or time of day. Its systems are used to track floods, wildfires, storms, and longer-term environmental changes, providing actionable data that supports climate resilience and recovery efforts.[110]

Founded in 2014, ICEYE initially focused on monitoring glacier melting but has since expanded into a wider range of applications, including defence and security, without fundamentally changing its underlying platform. Its competitive advantage comes from sending cheaper satellites to space that still provide constant monitoring, independent of the external conditions. Several European governments have recently reported having partnered with ICEYE, showing significant demand from the defence sector.[111] ICEYE has had 17 rounds of funding, valuing the company by the end of 2025 at 2.4 billion euros.[112]

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[107] Pollutec, "Critical Raw Materials: The EU Selects 60 Strategic Projects to Safeguard and Extend Its Supply Base," Pollutec Learn & Connect, November 20, 2025, https://learnandconnect.pollutec.com/en/critical-raw-materials-the-eu-selects-60-strategic-projects-to-safeguard-and-extend-its-supply-base-2/.
[108] Allianz Research, "Critical Raw Materials: Is Europe Ready to Go back to the Future?," *Allianz Research* (Allianz, August 1, 2023), https://www.allianz.com/content/dam/onemarketing/azcom/Allianz_com/economic-research/publications/specials/en/2023/august/01_08_2023-Critical-Raw-Materials.pdf.
[109] ICEYE, "ICEYE Recognized in the World's 50 Most Innovative Companies List by Fast Company," ICEYE, March 18, 2025, https://www.iceye.com/newsroom/press-releases/iceye-recognized-as-one-of-the-worlds-50-most-innovative-companies-by-fast-company.
[110] Ibid.
[111] Carter Palmer, "ICEYE Growing in Popularity as Europe Seeks to Improve Space-Based Capability," Govexec Space Project, January 27, 2026, https://spaceproject.govexec.com/defense/2026/01/iceye-growing-popularity-europe-seeks-improve-space-based-capability/410972/.
[112] ICEYE, "ICEYE and General Catalyst Partner to Redefine Space-Based Intelligence in Europe," ICEYE, December 5, 2025,

Current Financial Challenges (2025-2026): In 2025, ICEYE exceeded its target of raising 250 million euros and thus does not currently seem to be facing any need of cash, nor consider therefore going for IPO. Furthermore, its partnership with Rheinmetall in December 2025 allowed it to take on a 1.7 billion euro contract with the German Armed Forces, showing a strong demand for its products, which was highlighted again in Davos at the 2026 World Economic Forum.[113]

Strong demand following Russia's full-scale invasion of Ukraine, combined with multi-year government contracts, has provided ICEYE with stable revenue visibility. However, scaling a sovereign SAR constellation remains capital-intensive, requiring continued investment in satellite production, innovation and data infrastructure. ICEYE's challenge moving forward is to balance defence-driven demand with expansion in commercial markets such as insurance and climate risk analytics, while maintaining the financial flexibility needed to support global constellation growth.

Future Plans and Strategic Expansion:
ICEYE's growth trajectory is increasingly shaped by rising sovereign demand for space-based reconnaissance and situational awareness, particularly following the war in Ukraine and Europe's renewed focus on strategic autonomy. Partnerships with defence actors and long-term government contracts are likely to support further expansion of its SAR constellation and intelligence capabilities while seeking to preserve its position in the civilian markets. By the end of 2025, the company had launched 62 satellites after a decade of being active and it announced that it would want to ramp up production significantly to average 1 satellite per week in 2026, reflecting accelerating demand and capital-intensive nature of the enterprise.[114]

Relevance from an investment and policy perspective: From an investment perspective, ICEYE is relevant because it demonstrates the viability of a dual-use business model based on non-cyclical, multi-market demand that spans climate, insurance and defence actors. Rather than relying on a single end market, the company's model shows how climate-tech startups can attract defence demand as an upside while retaining a strong civilian and environmental relevance.

From a policy standpoint, ICEYE highlights the growing importance of space-based Earth observation as critical infrastructure for both climate adaptation and security. Its capabilities support disaster preparedness, environmental monitoring, and sovereign intelligence capacity, aligning with European objectives of strategic autonomy and data sovereignty. As climate risks intensify and geopolitical tensions persist, satellite-based situational awareness is likely to become an essential component of resilience planning and security governance

2) Vulcan Energy Resources

Vulcan Energy Resources is a German-Australian company that operates in the Upper Rhine Valley and was created in 2018. It focuses on extracting battery-quality lithium from geothermal brine. This provides a sustainable, carbon-neutral alternative to conventional lithium mining, which is vital for

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https://www.iceye.com/newsroom/press-releases/iceye-and-general-catalyst-partner-to-redefine-space-based-intelligence-in-europe.

[113] Seraphim Space Investment Trust, "ICEYE and Rheinmetall Sign €1.7 Billion Contract," London Stock Exchange News Service, 2025, https://www.londonstockexchange.com/news-article/SSIT/iceye-and-rheinmetall-sign-1-7-billion-contract/17381606.

[114] Anne Sraders, "Satellite Startup Iceye Hits €2.4bn Valuation as It Zeroes in on Defence," *Sifted*, December 5, 2025, <Satellite startup Iceye hits €2.4bn valuation as it zeroes in on defence>.

the European battery industry.[115] This would be crucial for the European Electric Vehicle (EV) industry as well as strategic sectors such as defence that currently have to rely heavily on imports for critical materials and batteries.

Vulcan Energy's Final Investment Decision on its Lionheart project in the Upper Rhine highlights the growing strategic relevance of low-carbon critical raw materials in Europe.[116] Backed by a €2.2 billion financing package combining commercial banks, EU public finance, and industrial partners, the project shows how climate-aligned lithium extraction can reach credible scale while supporting European supply-chain resilience and reducing execution risk for investors.[117]

Vulcan fits well into the regulatory framework of the EU as its operating model connects to a wider push for decarbonisation. Furthermore, its Lionheart project is designated as an EU "Strategic Project" under emerging CRM initiatives, which illustrates how upstream climate technologies are increasingly treated as assets of strategic importance, with relevance for both the green transition and broader security considerations.[118]

Current Financial Challenges (2025-2026):

Although the €2 billion Lionheart marks an important step in EU's CRM projects, it comes with its financing complexity that involves exposing investors to scale-up and delivery risks due to the multi-year construction timeline before the production begins.[119] As of December 2025, despite securing €2.2 billion in financing, Vulcan's €603 million equity raise and past dilution highlighted the capital intensity of the project, increasing investor sensitivity to execution performance and the risk of further funding needs.[120]

Future Plans and Strategic Expansion:

Vulcan Energy has entered the execution phase of its Lionheart project following the Final Investment Decision and the start of construction in late 2025. The project targets annual production of 24,000 tonnes of lithium hydroxide (sufficient for roughly 500,000 electric vehicle batteries) alongside renewable power and heat generation for local use.[121] Recent milestones, including an offtake agreement with Glencore, secured construction permits, and key engineering and supply contracts, position Vulcan to establish an integrated domestic lithium supply chain.[122] As construction progresses toward targeted production later in the decade, the company aims to play a central role in strengthening Europe's battery value chain and reducing dependence on imported raw materials.

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[115] Inno, "3 InnoEnergy Portfolio Companies Recognised as Strategic Projects under EU's Critical Raw Materials Act," Inno Energy, April 10, 2025, https://innoenergy.com/news-resources/3-innoenergy-portfolio-companies-recognised-as-strategic-projects-under-eus-critical-raw-materials-act/.

[116] Inno, "InnoEnergy Portfolio Vulcan Reaches Landmark FID for Lionheart Project," Inno Energy, December 8, 2025, https://innoenergy.com/news-resources/innoenergy-portfolio-vulcan-reaches-landmark-fid-for-lionheart-project/.

[117] Ibid.

[118] Ibid.

[119] European Investment Bank. "Germany: Vulcan Energy Secures €250 Million EIB Financing for Landmark Lithium Project." Accessed February 16, 2026. https://www.eib.org/en/press/all/2025-486-vulcan-energy-secures-eur250-million-eib-financing-for-landmark-lithium-project-in-germany.

[120] Simply Wall St, "Why Vulcan Energy Resources (ASX:VUL) Is Down 31.8% After Securing Full Phase One Project Funding – And What's Next," *Simply Wall St*, December 4, 2025.

[121] Vulcan Energy Resources Limited, "Quarterly Activities Report for the Period Ending 31 December 2025," *EQS News*, February 3, 2026.

[122] Vulcan Energy Resources Limited, "Quarterly Activities Report."

Relevance from an investment and policy perspective: From an investment perspective, Vulcan highlights the attractiveness of upstream climate technologies that benefit from structurally durable demand driven by electrification, energy storage, and industrial decarbonisation. By positioning itself within the battery supply chain, the company captures value at a critical bottleneck of the energy transition while reducing Europe's exposure to volatile global raw material markets.

Its development also illustrates the growing role of blended finance and long-term offtake agreements in de-risking capital-intensive climate infrastructure. As electrification accelerates and supply security becomes a strategic priority, projects such as Lionheart demonstrate how critical raw materials can evolve from commodity inputs into strategic assets that underpin the continent's industrial competitiveness and resilience.

3) Stegra (formerly H2 Green Steel)

In September 2023, this Swedish industrial decarbonization start-up secured Europe's largest private placement of the year at €1.5 billion, demonstrating growing institutional appetite for capital-intensive climate technologies.[123] Founded in 2020 and rebranded to "Stegra" in September 2024, the company is constructing the world's first large-scale green steel plant in Boden, northern Sweden.[124] The facility aims to produce steel with up to 95% lower CO2 emissions compared to traditional blast furnace technology.[125]

The company's production process replaces coking coal with green hydrogen produced by Europe's largest electrolyzer, powered entirely by renewable electricity from Sweden's hydropower grid. Steel production currently accounts for approximately 8% of global greenhouse gas emissions, making decarbonization of this sector a critical priority under the EU's *Clean Industrial Deal* and *Net-Zero Industry Act*.[126] Stegra's plant is designed to produce 5 million tons of green steel annually when fully operational, with initial production targeting 2026.[127]

Funding structure and public de-risking: Stegra exemplifies the blended finance model identified as essential for capital-intensive climate tech in this report. The company has secured €6.5 billion in total financing, comprising roughly €2.1 billion in equity and €4.2 billion in debt financing.[128] Critically, the project benefited from €250 million in grants from the EU Innovation Fund, alongside €1.2 billion in green credit guarantees from Sweden's National Debt Office (*Riksgälden*) and €1.2 billion in export credit cover.[129] This demonstrates how EU-level de-risking instruments successfully "crowd-in" private capital for first-of-a-kind industrial projects.

The company's investor base includes institutional climate-focused funds (Microsoft Climate Innovation Fund, Just Climate), sovereign wealth funds (GIC, Temasek), and the European

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[123] Stegra, "H2 Green Steel Raises €1.5 Billion in Equity for Green Steel Plant," Stegra, September 7, 2023, https://stegra.com/news-and-stories/h2-green-steel-raises-15-billion-in-equity-to-build-the-worlds-first-green-steel-plant.
[124] Stegra, "Decarbonizing at Scale," Stegra, September 12, 2024, https://stegra.com/.
[125] Stegra, "Green Steel," Stegra, 2025, https://stegra.com/green-steel.
[126] International Energy Agency, "Iron and Steel Technology Roadmap," IEA, October 8, 2020, https://www.iea.org/reports/iron-and-steel-technology-roadmap.
[127] Stegra, "Our Boden Plant," Stegra, 2025, https://stegra.com/the-boden-plant.
[128] Stegra, "H2 Green Steel Raises More than €4 Billion in Debt Financing," Stegra, January 22, 2024, https://stegra.com/news-and-stories/h2-green-steel-raises-more-than-4-billion-in-debt-financing-for-the-worlds-first-large-scale-green-steel-plant.
[129] Riksgälden, "Green Credit Guarantee for a Loan to H2 Green Steel," Riksgälden, December 22, 2023, https://www.riksgalden.se/en/press-and-publications/press-releases-and-news/news/2023/green-credit-guarantee-for-a-loan-to-h2-green-steel/.

Investment Bank (EIB), alongside over 20 international commercial banks.[130] Strategic industrial investors include Schaeffler and Hitachi Energy, while substantial offtake agreements have been secured with IKEA (Ingka Group) and multiple automotive OEMs.[131]

Current Financial Challenges (2025–2026): Despite its historic initial capital raises, Stegra is currently navigating severe financial headwinds typical of first-of-a-kind mega-projects. As of early 2026, the company has been actively raising an additional €1 billion to €1.5 billion in equity and debt to bridge a critical funding gap.[132] This liquidity shortfall is driven by several compounding factors. First, external partners and state-owned entities were initially slated to build essential logistics infrastructure, specifically the railway and Luleå port expansions. Bureaucratic delays forced Stegra to bring these massive, capital-intensive projects in-house to protect their production timeline. Second, general construction inflation has driven up the CAPEX of the Boden site, while friction with the Swedish government has delayed the disbursement of a €150 million grant previously approved by the EU Commission. Finally, the 2025 bankruptcy of Swedish battery manufacturer Northvolt—which shared a similar blended-finance model and early backers—has chilled the Nordic green industrial investment climate. In response, Stegra appointed a new CFO in February 2026 to close this defensive funding round, relying heavily on strong commitments from existing backers like Hy24 and Altor to maintain momentum.[133]

Future Plans and Strategic Expansion: Stegra's Boden plant is currently targeting initial operations in late 2026 or 2027, with the ultimate goal of producing 5 million tons of green steel annually by 2030. However, the company's global expansion strategy involves a significant operational pivot. Rather than replicating fully integrated, end-to-end steel mills worldwide, Stegra is actively evaluating sites in Portugal, Brazil, and Canada to act as upstream suppliers.[134] Benefiting from abundant renewable energy and high-quality iron ore in these regions, future Stegra facilities will likely focus solely on producing green hydrogen and Direct Reduced Iron (DRI) or Hot Briquetted Iron (HBI). This "green iron" would then be exported to traditional European steelmakers, allowing legacy producers to decarbonize the most energy-intensive step of their value chain without Stegra having to construct complex, €6-billion full-scale steel mills in every new geographic market.

Relevance from an investment and policy perspective: Stegra illustrates three structural dynamics central to EU climate tech financing. First, it confirms the capital intensity mismatch identified in Section 5: the company required €6.5 billion and nearly seven years from founding to commercial

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[130] Stegra, "H2 Green Steel Raises More than €4 Billion in Debt Financing," Stegra, January 22, 2024, https://stegra.com/news-and-stories/h2-green-steel-raises-more-than-4-billion-in-debt-financing-for-the-worlds-first-large-scale-green-steel-plant.

[131] Ingka, "Ingka Group (IKEA) and H2 Green Steel Sign Agreement for the Supply of Green Steel across Its Warehouse Operations," Ingka Group, September 13, 2023, https://www.ingka.com/newsroom/ingka-group-ikea-and-h2-green-steel-sign-agreement-for-the-supply-of-green-steel-across-its-warehouse-operations/.

[132] Siôn Lawrence-Geschwindt, "Green Steel Startup Stegra Downplays Insolvency Fears as It Seeks €1 Billion: 'Investors Are Firmly Committed,'" Impact Loop, October 13, 2025, https://www.impactloop.com/artikel/stegra-races-to-raise-eur1bn-as-it-downplays-insolvency-fears.

[133] Reuters, "Green Steel Startup Stegra Appoints Holm as CFO in Bid to Raise More Funds," ET Infra (Reuters, February 6, 2026), https://infra.economictimes.indiatimes.com/news/construction/swedish-green-steel-startup-stegra-appoints-markus-holm-as-cfo-amid-funding-challenges/127970958.

[134] Aidan Lea, "Sweden's H2 Green Steel Rebrands, Eyes Portugal Site," Argus (Argus Media, September 12, 2024), https://www.argusmedia.com/en/news-and-insights/latest-market-news/2607500-sweden-s-h2-green-steel-rebrands-eyes-portugal-site.

scale, timelines and ticket sizes far beyond conventional VC models. Second, it validates the "missing middle" thesis: without EU Innovation Fund support and national guarantees, private investors would likely not have mobilized at this scale for an unproven industrial technology. Third, the project demonstrates that when public de-risking is combined with strong industrial offtake and regulatory tailwinds, even highly capital-intensive climate technologies can achieve bankability.

III - Challenges and Opportunities for Climate Tech Start-Ups

A - SWOT Analysis for EU Climate-Tech Start-Ups

Strengths (structural advantages for Start-ups)	Weaknesses (structural constraints for Start-ups)
Structured Public Support Ecosystem: Permanent institutional architecture (EIF, EIB, national promotional banks) provides anchor LP support, blended finance tools, and counter-cyclical funding that reduces early-stage risk.[135]	**Fragmented Single Market**: 27 member states with divergent languages, currencies, and regulatory implementations create high transaction costs, slow cross-border scaling, and prevent the emergence of continental champions.
Dense Innovation Pipeline: Europe produces substantial technical talent and deep-tech start-ups, creating abundant early-stage deal flow at relatively lower valuations than in the US.[136]	**Bank-Dominated Financial System**: Banks are structurally ill-suited to finance pre-profit start-ups due to: (1) collateral requirements incompatible with intangible assets, (2) risk models unattuned to knowledge-intensive ventures, (3) maturity/debt-servicing mismatches, and (4) capital regulations that reduce returns on high-risk exposures.[137]
Reduced Late-Stage Competition: Scarcity of domestic mega-funds creates less competitive bidding pressure on growth-stage assets for investors with international capital access.[138]	**Missing Middle Funding Gap**: Critical shortage of growth capital (Series B+) for industrial-scale deployment, creating a persistent financing cliff between pilot projects and commercial scale.[139]
Emerging demand: market demand for clean and emission reducing products, both on consumer and national scales	**Small Private Capital Pools**: European pension funds and insurers hold €11.9 trillion in assets vs. €42.5 trillion in the US, resulting in smaller fund sizes, limited late-stage vehicles, and dependence on public capital.[140]

[135] See Footnote 19
[136] See Footnote 1
[137] See Footnote 67
[138] See Footnote 75
[139] See Footnote 45
[140] See Footnote 55

	Weak Exit Environment: Fragmented IPO markets across member states with lower liquidity and valuations compared to US exchanges, reducing expected returns and incentivizing start-ups to relocate.[141]
	Capital Intensity Mismatch: Climate-tech sectors (CCUS, green steel, hydrogen, clean materials) require substantially higher capex and longer commercialization timelines (~7 years) than standard VC fund horizons, creating structural tension with existing investor models.[142]

During our interviews experts consistently emphasized that Europe's **institutional support** infrastructure cannot fully compensate for its **structural financial system weaknesses**, particularly the bank-dominated model and fragmented markets. While public de-risking helps at early stages, the "missing middle" financing gap remains the critical bottleneck preventing European climate-tech start-ups from scaling to industrial deployment.

B - Opportunities and Threats

Opportunities (current dynamics favouring Startups)	Threats (current dynamics constraining Startups)
Regulatory stability as a long-term advantage: Stable and predictable EU climate regulation reduces policy risk for long-cycle technologies.	**Capital-intensive climate-tech sectors** (CCUS, green steel, hydrogen) require large early-stage funding rounds (often >$25m) and long commercialization timelines[143]. Novelty of much of the sectors and technology required further increases initial (research) costs.
Expansion of **venture debt and hybrid financing**: Venture debt and counter-guarantees reduce dilution and extend runway.	**Scale-up financing gap** between VC and debt: Funding cliff persists at industrial scale.

[141] See Footnote 42
[142] See Footnote 2
[143] Piotr Rubaj, "Competitiveness and Development of EU Countries in the Face of Global Economic, Social and Environmental Challenges," *European Research Studies Journal* 28, no. 2 (May 10, 2025): 22–37, https://ersj.eu/journal/3962/download/Competitiveness+and+Development+of+EU+Countries+in+the+Face+of+Global+Economic+Social+and+Environmental+Challenges.pdf.

EU **climate-neutral targets and innovation programmes** (e.g. Green Deal, JOULE, Horizon Europe) provide a clear innovation pathway, supporting a sustained increase in climate-tech innovation and patent filings (>20% annual growth)[144].	IP disputes and high costs of patent sharing slow cross-border diffusion of solar, wind, biomass and marine-energy technologies within the EU[145].
Public grants, export-credit guarantees and loan-guarantee schemes reduce debt risk and enable large-scale, capital-intensive climate-tech projects.	Grid constraints, uneven smart-grid maturity and fragmented national regulations delay renewable-energy connection and full utilization of output.

During our interviews, one main opportunity for European startups was stressed: regulatory stability combined with public de-risking mechanisms makes the EU a credible long-term scaling environment for climate-tech startups; **however, this potential is constrained by a persistent scale-up financing gap**, driven by investor risk aversion and structural limitations of bank-based financing models.

6) VCs: Investment and Funding Climate

I - Benchmark of Actors

Most active impact VC in 2025[146]

Name	Country	Focus	Nb deals 2025	Total deals participated (~$)
EIC Fund	EU	European Innovation Council	77	4.1 bn
Industrifonden	Sweden	Clean tech / deep tech	17	1.8 bn
Giant Ventures	UK	Seed & climate / inclusion	17	1.1 bn
Chalmers Ventures	Sweden	Early-stage science & climate spinouts	14	198 m
Norrsken VC	Sweden	Climate & tech innovation	13	4.9 bn
LUMO Labs	Netherlands	Article 9 VC — AI, climate, heavy industry	13	49.8 m
Asterion Ventures	France	Early-stage climate / deep	11	250 m

[144] Ibid.
[145] Ibid.
[146] Siôn Lawrence-Geschwindt, "Meet Europe's 30 Most Active Impact VCs in 2025," Impact Loop, December 22, 2025, https://www.impactloop.com/artikel/30-most-active-european-impact-vcs-in-2025-part-of-usd28bn-in-deals?paid=true.

		tech		
World Fund	Germany	Climate tech	11	931 m
Voima Ventures	Finland	Industrial & sustainable tech	11	160 m
Ananda Impact Ventures	Germany	Impact VC, climate & digital	10	410 m
Extantia Capital	Germany	Climate tech & emission reduction	10	336 m
Ada Ventures	UK	Breakthrough & deeptech (incl climate)	10	279 m
Vireo Ventures	Germany	Early-stage energy, mobility, climate solutions	10	90 m
AENU	Germany	Multi-stage climate & social impact	9	560 m
Inclimo Climate Tech Fund	Spain	Early-stage climate tech	9	36.0 m
Planet A Ventures	Germany	Early-stage systemic tech	8	474 m
Future Energy Ventures	Germany	Energy transformation & climate solutions	7	615 m
Nysnø Climate Investments	Norway	Long-term climate impact portfolio	7	566 m
Pale Blue Dot	Sweden	Seed climate tech	6	295 m
Contrarian Ventures	Lituania	Early industrial & climate tech	6	285 m
Faber	Portugal	Science & tech with climate action	6	268 m
The Footprint Firm	Denmark	Sustainability advisory & impact investments	6	77.7 m
Identity Ventures	Germany	Early-stage inclusive impact VC	6	34.6 m
SHIFT Invest	Netherlands	Early-stage clean tech, food & ag	6	184 m

II - Challenges and Opportunities for VCs in Europe

Strengths (structural advantages for VCs)	Weaknesses (structural constraints for VCs)

Strong regulatory tailwind: EU Green Deal, ETS, industrial policy create long-term demand visibility and regulatory pull for climate technologies.	**Fragmented single market**: regulatory, fiscal and linguistic barriers increase scaling costs and complicate pan-European investment theses.
Powerful public anchor LP ecosystem (EIF, EIB, national development banks): downside protection, counter-cyclical funding, early-stage risk sharing.	**Bank-dominated financial system:** structural under-allocation to equity, shallow late-stage rounds, dependence on subsidies.
Lower competition at late stage due to scarcity of domestic mega-funds: reduced bidding pressure on select assets.	**Limited private capital pools**: low VC exposure of pension funds and insurers resulting in smaller fund sizes and persistent scale-up funding gap.
Dense deep-tech and research pipeline: strong deal flow at lower entry valuations vs. US.	**Weak exit environment**: fragmented IPO markets, lower liquidity and valuations → reduced expected returns.
Ability to arbitrage international capital advantage for VCs capable of syndicating global investors.	**Loss of London as global VC hub**: remaining hubs lack critical mass and global gravitational pull.

Opportunities (current dynamics favouring VCs)	Threats (current dynamics constraining VCs)
Structural demand shift toward energy security, electrification, grid resilience and industrial decarbonisation : durable investment theses.	**Declining climate VC investment and fundraising momentum** : widening capital gap vs. US and Asia.
Market shift toward mature growth rounds (Series B+) aligns with investors able to structure large tickets and complex financing.	**Capital concentration around existing "champions" :** limits new entrant pipeline and reduces early-stage dynamism.
Rising electricity demand (AI, data centres, electrification): creates non-cyclical demand for climate infrastructure.	**Global competition, supply chain weaknesses** and higher production costs erode Europe's attractiveness for deployment.
Institutional LP demand for climate exposure supports specialised funds and structured products.	**Policy and macro volatility** (trade tensions, US policy shifts) increase investment uncertainty.
Large public capital deployment capacity (EIB €1tn to 2030) supports project finance and scale-up ecosystem.	**Dependence on public policy** slows execution and reduces ecosystem agility vs. market-driven regions.

7) Banks: Benchmark Bank Support for Climate Tech

Europe faces a significant investment gap in cleantech across the startup, scale-up, and growth phases. This financing shortfall constrains the ability of innovative companies to move from development to deployment and scale within Europe. As a result, many founders increasingly feel forced to relocate to the United States, where access to risk capital and growth financing is more readily available, raising concerns about Europe's long-term innovation capacity.

In this context, the role of banks has become increasingly critical. This section provides an overview of how HSBC's peers are currently supporting the climate tech ecosystem.

A - BNP Paribas

BNP Paribas has set a target to deploy EUR 200 billion in financing for the low-carbon transition between 2022 and 2025. As of late 2024, the Group has already deployed EUR 179 billion. This funding includes traditional loans and bonds, as well as private placements, strategic advice, and IPO support[147].

Key elements of their strategy:

- **Integrated Support:** They offer a "full lifecycle" model, positioning sustainable finance as a core pillar across corporate lending, capital markets, advisory, and asset management.
- **Dedicated Expertise:** Their Low-Carbon Transition Group consists of specialized bankers focused on emerging value chains like hydrogen, energy storage, batteries, and carbon-efficient industrial solutions. This group primarily helps large, established corporate clients pivot their business models.
- **Advisory & Networking:** Beyond lending, the bank acts as a strategic partner, advising innovative companies on capital raising to find the next "green unicorn".
- **Growth Capital & Solutions (GC&S):** This team is a one-stop-shop specifically designed to empower European growth companies. By combining commercial banking with specialized Corporate & Institutional Banking (CIB) services, they provide financing solutions such as venture debt and dynamic RCF products that climate tech firms need as they scale. (https://cib.bnpparibas/whats-next-for-innovative-companies-in-europe/)
- **Direct Equity Investment:** Unlike many of peer banks, BNP Paribas takes direct equity stakes in climate tech through dedicated venture vehicles like the Solar Impulse Venture Fund (BNPP SIVF). Launched in 2022 with €172m in capital, the fund targets 15 high-potential startups across Europe and North America that merge profitability with measurable environmental impact[148].

[147] BNP Paribas, "BNP Paribas Supporting Its Clients' Transition," *BNP Paribas* (BNP Paribas, 2024), https://cdn-group.bnpparibas.com/uploads/file/bnpparibas_supporting_its_clients_transition_2024_csr_achievements.pdf.

[148] Claire Thibeau, "BNP Paribas Solar Impulse Venture Fund Reaches Final Close, Raising €172m," BNP Paribas Asset Management (BNP Paribas, January 23, 2025), https://www.bnpparibas-am.com/en/press/mediaroom-en-bnp-paribas-solar-impulse-venture-fund-reaches-final-close-raising-e172m/.

Current portfolio examples:

Portfolio Company	Country	Technology / Focus	Strategic Relevance
NatureMetrics	UK	Biodiversity eDNA Monitoring	Nature Tech and biodiversity metrics
Phenix	France	Circular Economy Platform	Waste reduction and food recovery
Axioma	France	Agricultural Bio-solutions	Hydric stress resilience in agritech
Hello Watt	France	Residential Energy Renovation	Consumer-facing energy efficiency
Chemix	US	AI-powered Battery Design	Deep tech acceleration in mobility

B - Société Générale

Another major player in Europe's sustainability transition is Société Générale, which has recently committed to a new target for 2030: deploying EUR 500 billion in sustainable finance. This commitment covers financing, advisory services, and investments in sustainable bonds[149].

A core pillar of Société Générale's sustainability strategy is investment in innovative climate solutions through institutional partnerships:

(i) REED

Société Générale has established a dedicated energy-transition investment platform, REED – Société Générale Group, through a majority stake in Reed Management. The bank acts as an anchor investor, committing at least EUR 250 million to REED's first fund, which focuses on equity investments in emerging European energy-transition and climate-tech companies. REED's mission is to identify companies that have achieved initial success and provide them with the capacity to structure and scale into European and global champions.

The REED portfolio highlights the bank's focus on essential industrial sectors[150]:

Sector	Key Portfolio Initiatives	Strategic Asset Example
Energy	Renewable aggregation and management	**E6**: Integrated platform for renewable project development
Mobility	Heavy-duty road freight electrification	**Evergreen**: EV charging infrastructure for E-trucks
Water	Wastewater epidemiology and monitoring	**IAGE**: Quantitative molecular biology for water pathogens

[149] Société Générale, "Societe Generale Steps up Its Efforts Supporting the Environmental Transition," Société Générale, November 22, 2024, https://www.societegenerale.com/en/news/press-release/pr-societe-generale-steps-its-efforts-supporting-environmental-transition.

[150] REED, "Portfolio," Reed Management (REED Societe Generale Group, 2025), https://www.reed-management.com/en/portfolio.

Circular Economy	Closing the plastic loop (PET recycling)	**Loop Europe**: Chemical recycling platform using depolymerization
Digital	Next-generation energy-efficient data centers	**Voltekko**: Modular, eco-responsible data centers for AI

(ii) EIT InnoEnergy

Société Générale's support for climate tech extends into the broader European innovation ecosystem through its strategic partnership and equity investment in EIT InnoEnergy. As Europe's leading innovation engine for sustainable energy startups, EIT InnoEnergy manages a portfolio of over 200 companies. This partnership gives Société Générale early visibility into emerging industrial leaders, positioning the bank as a preferred financing partner when these startups enter the capital-intensive scale-up phase.

(iii) European Investment Bank (EIB)

Société Générale has also partnered with the European Investment Bank (EIB) under a Memorandum of Cooperation aimed at closing the financing gap for early-stage cleantech manufacturers. Through a EUR 250 million guarantee envelope backed by the InvestEU programme, the EIB provides risk-sharing mechanisms that enable Société Générale to offer specialized working capital solutions to "Emerging Leaders" in the energy transition. These facilities, including advance payment guarantees and performance bonds, are critical for startups facing long production cycles and constrained access to liquidity.

Overall, Société Générale's approach emphasizes risk-sharing structures and public–private partnerships, particularly to support emerging cleantech manufacturers and scale-ups navigating financing gaps during the early stages of industrialisation[151].

C - J.P. Morgan

Unlike BNP Paribas, J.P. Morgan does not operate a dedicated climate-tech equity fund. Instead, their strategy in Europe is built around enabling liquidity through their broader Innovation Economy platform**.** This platform provides venture banking, capital markets access, and strategic advisory services to high-growth technology companies.

Their approach is highly dependent on the startup's maturity stage:

- **Early Stage:** Offers operating accounts, liquidity management, and card processing for companies from pre-seed to seed.
- **Growth Stage:** Provides venture debt, private capital raising, and international expansion tools for venture-funded companies.
- **Late Stage:** Focuses on IPO preparation, M&A advisory, and complex treasury management.

To make this model work in Europe, J.P. Morgan has institutionalized its support by appointing dedicated regional leadership, Kai-Christian Nerger, who is based in Germany and leads their Green Economy Banking efforts across the continent. By focusing on three core pillars, Green Economy Banking, Sustainable Solutions, and the Center for Carbon Transition, they aim to finance $1 trillion by 2030 to support climate initiatives globally.

[151] European Investment Bank, "EIB and Societe Generale to Collaborate on Financing for Fast-Growing Innovative Cleantech Companies," European Investment Bank Group, November 19, 2025, https://www.eib.org/en/press/news/eib-and-societe-generale-to-collaborate-on-financing-for-fast-growing-innovative-cleantech-companies.

D - Role of Public Partners

Public financing instruments play a critical role in signalling European cleantech investment opportunities to private capital. They act as essential capital sources, enabling climate tech companies to overcome commercialisation challenges, particularly at early and growth stages. For instance, EU public funding institutions deploy capital with embedded risk-mitigation mechanisms, either by providing credit directly to growth companies that may face difficulties accessing traditional bank financing, or by de-risking private sector lending through guarantees, often allowing financing to be provided at a lower cost[152].

Beyond risk-sharing, official institutions also play a key role in scaling European innovation. For example, the European Investment Bank (EIB), notably through the EIF, supports nearly half of European VC-backed startups each year and has channelled billions of euros into major VC funds. As highlighted by BNP Paribas CIB, partnerships with official institutions such as the EIB and the EIF enable banks to share risk when financing growth companies, thereby unlocking greater volumes of capital than would be possible independently[153].

8) Decision Matrix: Guide for Climate Tech Support

I - Regulations

From the perspective of regulations, the main considerations to take into account when looking into climate tech start-ups or VCs to support is the extent to which they conform with EU-level regulations or other relevant initiatives[154]. In the context of this report, regulatory compliance serves three main functions: (1) it helps demonstrate climate commitment; (2) it reduces compliance risk for both investors and companies or start-ups; and (3) it helps identify opportunities for products or investments in the context of demand created by the regulatory environment.

Hence, compliance with the following relevant regulations is an important criteria:

- Compliance with the **EU Green Deal** entails alignment with EU climate objectives (such as decarbonisation, industrial green transition, etc.) for both start-ups and VCs.
    - Start-ups should ensure that their products contribute to the green transition objectives under the Green Deal, and that the emissions produced in their operations (supply chain or production process) are compliant with EU regulations.
    - Similarly, it is important for VC funds' investment strategy to align with Green Deal targets, and that VCs help create competitive advantage in deal sourcing in relation to clean tech.
- For VC funds particularly, compliance with the **EU taxonomy** classification is important, and helps identify the amount of projects or portfolio investments which contribute to the green transition (or more specifically are targeted towards climate tech).

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[152] Cleantech Group, "Mobilising Private Finance to Scale European Cleantech," *Cleantech for Europe*, 2025, https://www.cleantechforeurope.com/report-2024-25/report-mobilizing-private-finance-to-scale-european-cleantech.

[153] BNP Paribas, "What's next for Innovative Companies in Europe?," BNP Paribas CIB, February 9, 2026, https://cib.bnpparibas/whats-next-for-innovative-companies-in-europe/.

[154] Nick Evans, Matthias Duwe, and Eike Velten, "Policy Consistency: What It Means, How to Measure It, and Links with Other Processes Considerations on the Implementation of the EU Climate Law Ecologic Institute," *Ecologic* (Ecologic Institute, May 2, 2023), https://www.ecologic.eu/sites/default/files/publication/2023/50095-Policy-consistency-EUCL-implementation-web.pdf.

- VCs may also need to report their activity under the **SFDR**, mostly aligning with the CSRD, and highlighting their environmental impacts. This reporting highlights the level of operational sophistication and maturity of VCs, as well as their environmental impact and targets.
- For **CSRD** compliance, relevant considerations include tracking of emissions (scopes 1, 2, and 3) of all actors, whether they conduct double materiality assessments, and the transparency and verifiability of their data collection for reporting purposes
    - It is also important to consider the extent to which start-ups align their products with CSRD requirements, and their supply chain offshoring or supplying decisions (i.e. whether they supply to CSRD-compliant companies).
    - On the VC side, it is important to consider the number of CSRD-compliant portfolio companies.
- Finally, there are various relevant **policy schemes** for different actors, including compliance with sector-specific regulations and directives.
    - The **Clean Industrial Deal** and **Net Zero Industry Act**, amongst others, provide a screening measure for the extent to which start-up products and activities or VC fund exposure support decarbonisation in energy intensive industries and sectors.
        - They can also help identify whether start-ups or VC funds can capture regulatory-created demand stemming from policy initiatives and schemes in the EU
    - It is useful to consider whether certain climate tech companies receive funding/non-financial support under the **Green Deal Industrial Plan**, to measure the role of further bank support.
    - Lastly, policy initiatives help evaluate the extent to which climate tech receives specialised support if it falls under a certain sector (ie. critical raw materials, renewable energy, construction and buildings, etc.)
        - This is outlined in more detail in the Start-up sectors section.

Strategic option	Elements of EU context	Value for HSBC
Track compliance	Employ EU metrics of regulatory/policy compliance[155]	Filters sustainability-focused activities and projects with standardised EU classifications and definitions; decreases risk and uncertainty
Track policy and regulatory developments	Track EU regulatory and policy framework amendments and developments	Filter strategic sectors and activities at EU level; decreases risk and uncertainty

[155] Nick Evans, Matthias Duwe, and Eike Velten, "Policy Consistency: What It Means, How to Measure It, and Links with Other Processes Considerations on the Implementation of the EU Climate Law Ecologic Institute," *Ecologic* (Ecologic Institute, May 2, 2023), https://www.ecologic.eu/sites/default/files/publication/2023/50095-Policy-consistency-EUCL-implementation-web.pdf.

	Track EU-level support (financial or not) provided to various sectors/actors	
Select relevant policy initiatives by sector or activity	EU regulations, directives, or initiatives for specific financial actors (VCs or others), specific climate tech sectors for climate tech demand or deployment	Narrow down activity and relevant environmental classifications
Platform for connecting financial clients and start-ups	CSRD and Clean Industrial Deal create demand for clean tech; SFDR help classify environmental impact Work with existing platforms such as cleantech for Europe to streamline projects	Aligns with climate transition support goals, helps in emission reduction targets; risk-sharing
Advising investors on climate strategy	Regulatory demand for investments in sustainable activity, and increasing amounts of potential climate tech projects	Aligns with existing corporate client relationships, as well as climate tech deals and expertise

II - Financing Schemes

Regarding financing or funding schemes, it is relevant to investigate where the climate tech start-up or VC fund benefits from funding or investment support from EU-level or other public schemes.

This includes two distinct facets:

(1) Profitability and feasibility of deployment of the climate tech under consideration;
(2) Necessity of further financial or other types of support.
    (a) For instance, several climate tech projects under the InvestEU, Horizon Europe, or the Innovation Fund receive sufficient funding and expertise from the EU to successfully deploy their projects
    (b) Others, such as those under the IPCEI, receive member state support and usually also appeal to outside support, networks, and expertise

Strategic option	Elements of EU context	Value for HSBC
Partner with initiatives and funds	Existing initiatives (Innovation Fund) may lack external partners, especially private financing	Helps connect HSBC to climate tech start-ups already receiving EU support (decreases risk)

Assess profitability of climate tech under EU initiatives	Support offered to climate tech at the EU level does not guarantee competitiveness or profitability	Connect with start-ups and relevant EU initiatives, aligns with networking strategy
Position as a private-sector partner in fixing EU cleantech scale-up gap[156]	EU scale-up and project deployment issues can be addressed through private sector support	Aligns with existing corporate client relationships, as well as climate tech deals and expertise

III - Competitive Environment

Strategic option	Elements of EU context	Value for HSBC
Focus on Growth / Series B-C climate tech	Market shifting toward fewer, larger, later-stage rounds.	Aligns with balance-sheet strength, risk structuring and large ticket capacity.
Act as international capital bridge	European funds lack mega-funds and global syndication capacity.	Differentiate via global LP network, cross-border syndication, co-investments.
Partner with public institutions	Public actors are anchor LPs and major risk mitigators in the EU ecosystem.	Risk-sharing, privileged deal flow, policy alignment.
Selective early-stage exposure via fund investments	Early-stage remains risky but critical for pipeline access.	Access to innovation without direct risk concentration.
Avoid broad generalist VC strategy	Fragmentation and capital scarcity penalise non-specialised players.	Prevent dilution of competitive advantage.
Update model and embrace risk	Financial system dominated by banks that are risk-averse.	Adapt the current tools to the needs of the VC ecosystem.

IV - Start-Ups

Closely related to the competitive environment for climate tech is the consideration of which sectors are most competitive or relevant to support in the EU. The sectors identified in this report were industry; (renewable) energy; dual-use tech (climate + military or defense technologies); built environment or construction; carbon capture, usage, or storage; and critical raw materials. Identifying

[156] See Footnote 55

the opportunities and challenges of climate tech in each sector is crucial in assessing the profitability and competitiveness of various projects.

Strategic option	Elements of EU context	Value for HSBC
Assess profitability and competitiveness of projects	Competitiveness of climate tech in sectorial context, and regulatory/financing background	Define risk and profitability of projects and climate tech
Connect private and public capital with sector-specific projects	Investment and funding constraints in many sectors, lack of access to private finance	Aligns with existing corporate client relationships, as well as climate tech deals and expertise; allows for risk-sharing
Private-sector partner	Partner with public sector (funding or non-financial support)	Aligns with existing corporate client relationships, as well as climate tech deals and expertise; allows for risk-sharing

FOCUS - Invest or not invest? Green and red flags in the decision-making at different stages of financing

Subsector : Battery production/recycling

Scope: EU
Context: capex heavy, regulated sector, face execution risk.
Purpose: decision-useful checklist to assess whether to provide (i) venture debt at early stages and (ii) growth/scale debt or structured financing at later stages in the battery production/recycling subsector
How to use: apply the must-not-fail gates first (binary). If gates pass, decide depending on the financing stage: proceed, proceed with mitigants, or decline.

1. **Must-not-fail gates:** these gates are non-negotiable. If any gate fails, the decision should be paused or declined, until fixed. Negative EBITDA is not automatically a fail at early stages, but repayment path + sponsor support + runway discipline must be strong.

Gates		What to verify	Typical fail signs
Integrity and compliance	Can legally and safely bank the company.	Clear ultimate beneficial ownership (UBO), KYC/AML file, sanctions	Hidden owners; inconsistent disclosures; major unresolved legal

		screening, litigation overview; basic governance and controls.	issues; repeated compliance incidents.
Verifiable reality (no story-only deals)	Key claims are backed by evidence.	Evidence of customers' contracts/invoices; product performance proof (tests/pilot results); third-party testing where relevant; consistent financial and KPI reporting, coherent data room.	Refusal to share basic proof; KPIs change frequently; selective or contradictory documentation; NDA used as blanket excuses.
Repayment path exists	There is a credible way the loan will be repaid or refinanced.	Early stage: likely via next equity round / improving unit economics / ARR growth. Later stage: operating cash flows supported by bankable contracts (feedstock and offtake).	Repayment relies on hope without credible next round; weak sponsor base; forecasts rely on "perfect execution" - overly optimistic forecasting without downside plan.
Use of proceeds creates value	Debt funds clear milestones (growth/scale); no structural cash leakage.	Budget tied to milestones: deployments, capex, working capital tied to signed contracts. => Defined milestones and timeline; use-of-funds consistent with business model.	Vague use of funds; mainly to cover past holes indefinitely; no milestone plan.
Green integrity and regulatory viability	Impact claims are credible and operations are licensable; no obvious greenwashing.	Clear impact KPIs and measurement plan; lifecycle thinking where relevant; supply-chain safeguards (minimum social/environmental standards).	Vague "net-zero" claims with no measurement; major ESG controversy without remediation plan.

2. Early-stage matrix (venture debt lens) - battery recycling

Assuming the must-not-fail gates pass, early-stagers often have negative EBITDA due to R&D, pilots and early industrialisation. Venture debt can still be rational if (i) equity sponsors are credible and likely to support follow-on rounds, (ii) post-financing runway is sufficient, and (iii) the facility is structured defensively (tranches, covenants, milestones). Traction is not only revenue: it is also validated process performance and credible pathways to both feedstock supply and offtake.

What to look at		What to verify	Green flags	Red flags
Customers and market traction	"Customers" as paying entities (not just pilots). Renewals/expansion as a sign of product-market fit.	Number of paying customers, types of contracts, renewals, concentration.	Paid contracts + renewals; customers expand usage/orders; diversified counterparties.	Pilot-heavy with little conversion; one customer dominates revenue
Unit economics	Even if EBITDA is negative now, is there a credible path to positive margins at scale?	Processing cost, expected revenue, CAC (cost to acquire a customer), gross margin, scalability, discount policy.	Clear improvement path; repeatable yields; costs trend down with scale.	Economics only work under optimistic prices/external conditions; unstable yields; no margin pathway
Technology validation	Does the process work outside lab and can it scale?	Pilot results; third-party tests; scalability plan.	Repeatable pilot outcomes; clear scale-up plan; strong safety controls.	Inconsistent results; unclear technology, safety incidents ignored.
Sponsor quality/capacity	Can they bridge delays or fund the next round if timelines slip?	Investor quality, fund size, follow-on history; governance.	Reputable VCs/strategics; follow-on credibility; active governance.	Weak syndicate, fatigue; debt used because equity is unavailable.
Grants / guarantees / de-risking package	Non-dilutive support can reduce credit risk.	Confirmed grants, public programmes, risk-sharing structures, insured performance guarantees, or strong strategic partner commitments.	Grants awarded; clear eligibility and disbursement timeline; reputable counterparties.	Uncertain expected grants; slow disbursement; grant conditions not understood.

Runway and liquidity	Check months of cash available after financing; must be robust given uncertainty.	Cash post-deal; downside plan; minimum cash needed to operate safely.	Ideally >= 12 months runway; credible cost savings in downside case.	< 9 months runway; plan assumes perfect execution; absence of downside plan.
Regulations	Sector (recycling) is regulated	Permit roadmap; site readiness; compliance; insurance; audits.	Clear permitting plan; compliance; insurance.	Permits uncertain; safety not clearly treated; unclear liabilities.
Execution capability	Check execution capabilities of the team. Can it deliver milestones on time with disciplined reporting and controls?	Key-person risk, hiring plan realism, reporting.	Milestones achieved; good reporting; credible hiring plan.	Leadership weak; key-person risk; chaotic.
Structure (tranches and covenants)	Match funding to milestones; transparency via reporting.	Milestone schedule: pilot -> FOAK -> commissioning; reporting; negative covenants on extra debt/leakage.	The company accepts milestones and reporting; data is consistent.	Funding not matching the milestones (ex: required all upfront).

Example: tozero (Germany)

Public information indicates tozero raised 11 million EUR in an oversubscribed seed round to build its first industrial deployment (FOAK) plant and scale production (investors cited publicly include NordicNinja and new investors; e.g., In-Q-Tel, Honda, JGC Group). Despite negative EBITDA, venture debt may still be considered at this stage if sponsor quality is strong, runway is sufficient, and the facility is based on milestones.

Comparison with late-stage investments (growth/scale debt)

Always assuming the must-not-fail gates pass, later-stage underwriting relies more on bankable cash-flows visibility (i.e., through contracts), liquidity buffers during scale up, execution readiness, and asset-backed structure (i.e., covenant/security packages). If EBITDA is still negative, it is important to check whether there is a credible path to cash flow generation and sufficient liquidity to

survive the delays. Checks on the business model / cash conversion mechanisms (how revenues are turned into cash - i.e., payments terms, margins) and metrics like the DSCR are crucial at this point.

Verkor (France) : late-stage financing success	**Northvolt (Sweden) : late-stage financing failure**
Verkor secured a layered, risk-shared financing package: in 2023 it announced >€2bn combining a Series C equity round (min 850m EUR), EIB debt support (600m EUR) and French subsidies (650m EUR, subject to approvals), and it made explicit that its plan was backed by established contracts with major ("tier-one") partners, including a long-term commercial partnership with Renault Group (anchor customer agreement that helps lenders feel there is real, committed demand). In May 2024, Verkor then closed a 1.3bn EUR green loan, for its Dunkirk gigafactory, with 19 banking entities (16 commercial banks and 3 public banks), including 961m EUR of senior loans of which 353m EUR was guaranteed by the French government (GPS). Plus EIB participation and public-bank instruments (bridge/subordinated loans), which again spreads and reduces risk for lenders.	Northvolt also secured very large-scale funding, but its financing story ultimately broke on execution and liquidity. In its own bankruptcy statement, Northvolt said its financial position was hit by rising capital costs, geopolitical instability, supply-chain disruptions, and shifts in market demand, and that this was compounded by significant internal challenges in ramping up production (some expected in a complex industry, others unforeseen). As ramp issues persisted, the commercial impact became visible: BMW cancelled a 2bn EUR battery cell supply contract, citing delivery timelines, which weakened demand visibility and confidence even more. Northvolt then sought protection via a Chapter 11 restructuring in the US to access about 245m USD in new liquidity, but this was a temporary bridge rather than a full solution. In March 2025, Northvolt said that despite liquidity support, it could not secure the financial conditions to continue and filed for bankruptcy in Sweden.

V - Banks

This comparison table outlines the climate-tech financing approaches of key global banks:

- BNP Paribas and Société Générale build integrated and institutional partnership models that directly deploy balance sheet and public risk capital.
- J.P. Morgan focuses on innovation-economy support that feeds into capital markets monetisation, especially for later-stage growth.
- HSBC uses its Innovation Banking platform to deploy tailored, non-dilutive debt across the entire venture lifecycle, while leveraging its vast international network to connect high-growth startups with corporate clients.

	HSBC	BNP Paribas	Société Générale	J.P. Morgan
Strategy	Bridges tech deployment between Europe, Asia, and the US	Largest EU integrated lender with a focus on regional transition	Engineering-led focus on complex hardware and mobility	Focus on late-stage growth, M&A, and US listings

Venture Product	HSBC Innovation Banking: High-volume venture debt (Seed to IPO)	Solar Impulse Fund: Strong, direct equity partnership for "labeled" startups	REED Management: Direct equity for "Industrial Champions" and green leasing	Green Economy Banking: Specialized advisory and debt for late-stage startups
Key Competitive Edge	Introduces startups directly to its global corporate client base	Massive lending capacity for large-scale EU infrastructure	Deep expertise in hydrogen and EV supply chains via EIT InnoEnergy	Access to US capital markets and NASDAQ/NYSE listings
Risk Appetite	Mitigates high-risk lending by using Non-Delivery Insurance, which protects the bank if a project fails to deliver its promised product	Leverages significant EU public-private risk sharing	Focuses on de-risking via asset-backed financing and leasing	Lower appetite for "first-of-a-kind" (FOAK) construction risk

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10) Appendix

I - Regulations

A - EU-level Regulations

- The **Fit for 55** package is a set of EU laws created with the aim to reduce EU emissions by 55% by 2030[157]. As maintaining innovation and competitiveness of EU industry is a key goal of the package, many of its rules and related social or climate funds are directly relevant to climate tech (notably in the sectors of energy, building, and transport).
    - It also includes a reform of the EU ETS, expanding to the building, transport and fuels, and aviation sectors.
    - The CBAM, adopted in 2023, was also proposed under the Fit for 55 package.
- The **EU Emissions Trading Scheme** (ETS) and **Carbon Border Adjustment Mechanism** (CBAM) are two sets of regulation featured under the Green Deal, establishing the EU carbon market. The ETS scheme requires polluters to pay for the greenhouse gases (GHG) they emit by having a limit on the GHG emissions allowed and a trading system for emission permits (a "cap-and-trade" system)[158]. It is a market-based carbon pricing scheme, with average values of 65€/ton of CO2 in 2024.
    - The CBAM extends this scheme to carbon-intensive goods produced abroad and entering the EU, requiring producers to pay depending on their carbon emissions[159].
    - The revenues from the ETS and CBAM directly enter the EU's green transition goals, and thus can be invested into innovation or deployment of low-carbon/clean technologies.
- The new **Multiannual Financial Framework** (MFF) concerning the 2028-2034 EU budget is also important as certain sections focus specifically on clean tech financing, notably in research and innovation[160]. The research goal of the MFF was established with a focus on the clean transition and decarbonisation.

B - Regulations for Start-Ups

- The **Clean Industrial Deal** aims to support EU industries to decarbonise, while maintaining low energy prices and creating jobs[161]. It focuses on energy-intensive industries as well as investing in clean tech to support decarbonisation and the green transition. It includes national clean tech procurement quotas as well as renewable targets and efficiency mandates for industries and member states. Additionally, it offers permits covering grids, energy storage, and renewables.
    - The **Clean Industrial Deal State Aid Framework** (CISAF) was adopted in 2025 to help member states implement the Clean Industrial Deal and develop clean energy,

[157] European Council, "Fit for 55," European Council, 2025, https://www.consilium.europa.eu/en/policies/fit-for-55/.
[158] European Commission, "About the EU ETS," Climate Action, 2024, https://climate.ec.europa.eu/eu-action/carbon-markets/about-eu-ets_en.
[159] European Commission, "Carbon Border Adjustment Mechanism," Taxation and Customs Union, 2025, https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en#sectoral-information.
[160] European Commission, "EU Budget 2028-2034," European Commission, 2026, https://commission.europa.eu/strategy-and-policy/eu-budget/long-term-eu-budget/eu-budget-2028-2034_en#factsheets.
[161] See Footnote 14

industrial decarbonisation, and clean technologies[162]. Rather than a regulation, it is a framework to support state-led efforts in clean tech development and deployment.

- The **Green Deal Industrial Plan**, established specifically to support the development of clean tech in the EU and enhance the region's competitiveness[163]. It aims to do this through a simplified regulatory environment, easier access to funding, investing in relevant skills, and opening trade for resilient supply chains.
    - The second pillar of easier and faster access to funding is especially relevant for climate tech start-ups, as it increases access to funding from member states and from EU-level funds (also increasing the amount of investments in this sector).
- The **Net Zero Industry Act** (NZIA) is a framework to promote the "competitiveness of EU industry and technologies crucial for decarbonisation"[164]. Hence, it is directly linked to supporting the manufacturing and research of clean technologies. Moreover, it aims to attract investments to and improve market access for clean tech in the EU, addressing barriers to production and funding faced by the sector.
    - The NZIA covers final output, components, and machinery of various climate tech, encompassing all kinds of renewable energy-related technologies (from solar, to heat pumps and hydrogen, to wind)
    - It also covers CO2 transport and use technologies and carbon capture and storage (CCS) technologies.
    - The regulation's focus on investment certainty provides easy and centralised access to investment information, both at the national and EU level and at the private level.
- The **Important Projects of Common European Interest** (IPCEI), run with national budgets from EU Member States, aim to fund beneficial projects at the EU-level that do not receive enough attention or budget from the EU or within Member States[165]. While most IPCEIs focus on the medical or energy sectors, a few are directly linked to the green transition, such as the four hydrogen IPCEI[166].
- Specialised policy schemes for specific sectors, including:
    - The **Critical Raw Materials Act**, concerning the sourcing and processing of critical raw materials[167];
    - The **Energy Performance of Buildings Directive**, aiming to decarbonise and decrease the energy consumption of the building sector[168];
    - The **FuelEU Maritime Regulation**, covering the need to decrease the emissions of the shipping sector[169];

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[162] European Commission, "Clean Industrial Deal State Aid Framework (CISAF)," Competition Policy, January 2026, https://competition-policy.ec.europa.eu/about/contribution-clean-just-and-competitive-transition/clean-industrial-deal-state-aid-framework-cisaf_en.
[163] See Footnote 15
[164] See Footnote 16
[165] European Commission, "Important Projects of Common European Interest (IPCEI)," Competition Policy, 2025, https://competition-policy.ec.europa.eu/state-aid/ipcei_en.
[166] European Commission, "Approved Integrated Important Projects of Common European Interest (IPCEI)," Competition Policy, 2025, https://competition-policy.ec.europa.eu/state-aid/ipcei/approved-ipceis_en.
[167] See Footnote 17
[168] See Footnote 18
[169] European Parliament and Council of the European Union, "Regulation (EU) 2023/1805 of the European Parliament and of the Council of 13 September 2023 on the Use of Renewable and Low-Carbon Fuels in Maritime Transport, and Amending Directive 2009/16/EC" (2023), https://eur-lex.europa.eu/eli/reg/2023/1805/oj/eng.

- The **ReFuelEU Aviation Regulation**, aiming to contribute to the competitiveness of the EU aviation sector and promote its sustainable development[170];
- The **CountEmissions EU Regulation**, reporting emissions for passenger and freight transport in a centralised manner[171];
- Etc.

C - Regulations for VCs

- The **EU Venture Capital Regulation** (EuVECA), which will be reviewed and updated in 2026[172]. Its aim is to regulate Alternative Investment Funds (AIF) and Venture Capital Firms while also aiding them in investing in a more targeted and controlled manner in the EU. It allows private equity and venture capital firms to market their funds across the EU more efficiently, giving them a permit (or "marketing passport") to operate in the EU without needing to comply with the AIF managers directive[173].
- The **Savings and Investment Union** (SIU), a European initiative aiming to deepen capital markets in the EU and addressing barriers to investors[174]. It enables funding and savings sources to be linked with "productive investments" at the EU level, especially in the climate and technology sectors[175]. Most of the investment needs focus on SMEs and innovative companies (including climate tech), connecting them to potential investors at all levels, from citizens to VCs. As of the end of 2025, the initiative also includes a market integration package and further clarification of equity investment.
    - The market integration package aims to provide a better developed, more competitive and more integrated EU capital markets, which would also increase the efficiency of VC funding[176].

II - Financing Schemes

A - EU-level Financing

[170] European Parliament and Council of the European Union, "Regulation (EU) 2023/2405 of the European Parliament and of the Council of 18 October 2023 on Ensuring a Level Playing Field for Sustainable Air Transport (ReFuelEU Aviation)" (2023), https://eur-lex.europa.eu/eli/reg/2023/2405/oj/eng.
[171] European Commission and Directorate-General for Mobility and Transport, "Commission Welcomes Political Agreement on New Rules to Harmonise Transport Emissions Calculations in the EU," Mobility and Transport, November 6, 2025, https://transport.ec.europa.eu/news-events/news/commission-welcomes-political-agreement-new-rules-harmonise-transport-emissions-calculations-eu-2025-11-06_en.
[172] European Commission, "Savings and Investments Union: A Strategy to Foster Citizens' Wealth and Economic Competitiveness in the EU," COM/2025/124 final § (2025), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52025DC0124.
[173] Invest Europe, "EuVECA | Invest Europe," VC fund regulation (Invest Europe, 2025), https://www.investeurope.eu/policy/key-policy-areas/euveca/; European Parliament and Council of the European Union, "Directive 2011/61/EU of the European Parliament and of the Council of 8 June 2011 on Alternative Investment Fund Managers and Amending Directives 2003/41/EC and 2009/65/EC and Regulations (EC) No 1060/2009 and (EU) No 1095/2010" (2011), https://eur-lex.europa.eu/eli/dir/2011/61/oj/eng.
[174] European Commission, "Savings and Investments Union: A Strategy to Foster Citizens' Wealth and Economic Competitiveness in the EU," COM/2025/124 final § (2025), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52025DC0124.
[175] European Commission, "Savings and Investments Union," Finance, 2025, https://finance.ec.europa.eu/regulation-and-supervision/savings-and-investments-union_en.
[176] European Commission, "Market Integration Package," Finance, December 4, 2025, https://finance.ec.europa.eu/publications/market-integration-package_en.

- **InvestEU**, an initiative launched in the Green Deal in 2021 and implemented mostly by the European Investment Fund.[177] Focused on increasing investments to projects linked to sustainability, innovation, and social concerns, it guarantees support to the energy, transport, digital and circular economy sectors. It also aids in mobilising private capital and risk-sharing guarantees for projects in the climate sector.[178]
    - The main sources of funding come from the European Investment Bank (EIB), EU budget guarantees, funding from national promotional banks, and connections to private investors.[179]
- **NextGenerationEU** and **REPowerEU**, funded as part of the Green Deal. It provides €275 billion in clean investments, and 42% of its funds are directly dedicated to the green transition and climate goals.[180]
- **Horizon Europe** is a research and innovation initiative to provide support and funding (mainly in the form of grants) to various important research projects and technology development.[181] As the programme's main targets are climate change, sustainability, competitiveness, and growth; at least 30% of the budget is dedicated to climate-related projects and technologies. Adopted as part of the Green Deal and Industrial Strategy, the budget for the initiative was decided as €95.5 billion over 2021-2027, but with the newest proposal for 2028-2034 aiming to double this sum.[182]
    - The missions linked to Horizon Europe (such as the climate-neutral cities or the climate adaptation missions) fund research and innovation often in the climate sector and help de-risk and deploy climate tech[183].
- The **European Hydrogen Bank** is a financing instrument established in 2022 as part of REPowerEU and the Net-Zero Industry Act. It specifically targets hydrogen value chains in the EU, with both a domestic and international action plan to promote renewable and competitive hydrogen value chains[184]. Most of its funding comes from the Innovation Fund, through allowances from the EU ETS programme.
- **Strategic Technologies for Europe Platform** (STEP), a funding initiative focused specifically on technologies in various sectors, including (but not limited to) climate technologies[185]. It targets the development and manufacturing phases of technologies to help with innovation and production, rather than deployment. "Clean and resource efficient technologies", encompassing all climate and clean tech, is one of STEP's four main target sectors[186]. Funding is provided by the European Innovation Council[187].

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[177] See Footnote 31
[178] Ibid.
[179] Ibid.
[180] European Union, "NextGenerationEU," EU, 2024, https://next-generation-eu.europa.eu/index_en.
[181] See Footnote 34
[182] See Footnote 35
[183] European Commission, "EU Missions in Horizon Europe," Research and innovation, January 2026, https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe/eu-missions-horizon-europe_en.
[184] European Commission, "European Hydrogen Bank," Energy, 2025, https://energy.ec.europa.eu/topics/eus-energy-system/hydrogen/european-hydrogen-bank_en.
[185] European Union, "Strategic Technologies for Europe Platform," EU, February 2026, https://strategic-technologies.europa.eu/index_en.
[186] European Union, "STEP Scope," Strategic Technologies for Europe Platform, December 2025, https://strategic-technologies.europa.eu/about/step-scope_en.
[187] European Innovation Council, "STEP Scale Up," European Innovation Council, November 26, 2024, https://eic.ec.europa.eu/eic-funding-opportunities/step-scale_en.

- The **Innovation Fund**, created under the EU ETS directive and integrated into the Clean Industrial Deal, provides €40 billion from 2020-2030 to the deployment of clean and innovative technologies[188]. The funding is acquired through the EU ETS system, and is allocated through competitive tenders and auctions to climate tech in the industry, renewable energy, energy storage, carbon capture, and mobility and buildings sectors. It is considered the main fund for the EU's climate policy, and as of 2025 was expanded to include new standards and sectors.
- The **LIFE Programme** is an initiative divided into four sub-programmes that each focus on an aspect of the green transition, with the most relevant ones for climate technologies being the "Climate change mitigation and adaptation" and the "Clean energy transition" pillars[189]. It has a budget of around €905 million to allocate to climate action overall, including clean tech. Funding is acquired through the Direct EU budget, and it provides grants for research and innovation, adaptation and mitigation, and governance projects[190].
- Lastly, the **Modernisation Fund** was created in 2018 as part of the EU ETS reform, and is active for the 2021-2030 period with around €23 billion in funding[191]. It supports modernisation and investments in renewable energy, energy efficiency, and decarbonisation in the 13 lower-income EU member states (mostly Central, Eastern, and Southern Europe). The Fund is mostly financed by EU ETS allowances and allocating it to relevant projects in the member states concerned by the fund.

B - Financing by EU Actors

- The **European Investment Bank**'s (EIB) climate finance targets and its climate bank roadmap, with the aim to lend for the green transition, specifically to scale climate technologies, innovation, and infrastructure[192]. To achieve this goal, the EIB provides green financing for direct borrowers and financial intermediaries through loans, venture debt, direct equity investments, and guarantees.
- The **European Investment Fund** (EIF) provides equity investments in VC funds using public or private capital, under climate action and sustainability criteria requiring at least 90% of the VC's revenue stemming from environmentally sound activities[193].
    - **The Cleantech Co-Investment Facility** is a €200 million equity co-investment instrument targeting companies contributing to the EU's objective to achieve climate neutrality by 2050. Launched in January 2024, deployed over a period of four years. Its purpose is to increase investment into innovative green technologies and business models that will have a meaningful impact on the climate transition, in line with the European Green Deal objectives[194].

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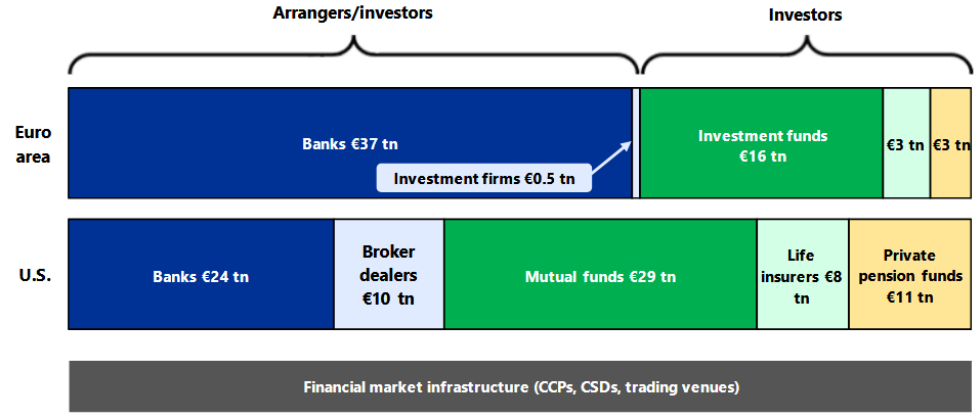
[188] See Footnote 33
[189] European Commission, "LIFE Climate Change Mitigation and Adaptation," Climate Action, 2025, https://climate.ec.europa.eu/eu-action/eu-funding-climate-action/life-climate-change-mitigation-and-adaptation_en.
[190] European Commission, "Climate Change Mitigation and Adaptation," European Climate, Infrastructure and Environment Executive Agency, 2025, https://cinea.ec.europa.eu/programmes/life/climate-change-mitigation-and-adaptation_en.
[191] European Commission, "Modernisation Fund," Climate Action, 2026, https://climate.ec.europa.eu/eu-action/eu-funding-climate-action/modernisation-fund_en.
[192] See Footnote 36
[193] See Footnote 37
[194] European Investment Fund, "Cleantech Co-Investment Facility," European Investment Fund, 2026, https://www.eif.org/products/all/cleantech-co-investment-facility.

        - The Facility will focus on supporting European SMEs and MidCaps, and specifically targets the "second equity gap" that companies often face when moving beyond start-up stage and into the growth phase of their lifecycle (TRL≥7)[195].
    - EIF has also increased its investment activity in climate and environmental impact funds and has committed close to EUR 1 billion in more than 25 VC, PE and PC (Private Credit) funds, effectively doubling the 2022 investment volumes, as well as a further EUR 1 billion in climate and infrastructure funds[196].
    - Possible objective : The EU has a scaleup financing gap. Europe is not short of innovative companies, but those that want to grow face serious difficulties in accessing capital at home. Typically, venture capital (VC) firms provide capital to risky startups[197].
- The **European Innovation Council** (EIC) is an initiative to support research and innovation in the EU[198]. The EIC fund provides financial support to projects and initiatives such as the STEP Scale Up and the EIC Accelerator amongst others[199]. The Accelerator provides direct equity investments into early-stage scale-ups, as well as scale-up grants and entrepreneurship advice.

III - Competitive Environment

A - Annexes from the IMF report[200]

Figure 5: Financial Market Structure

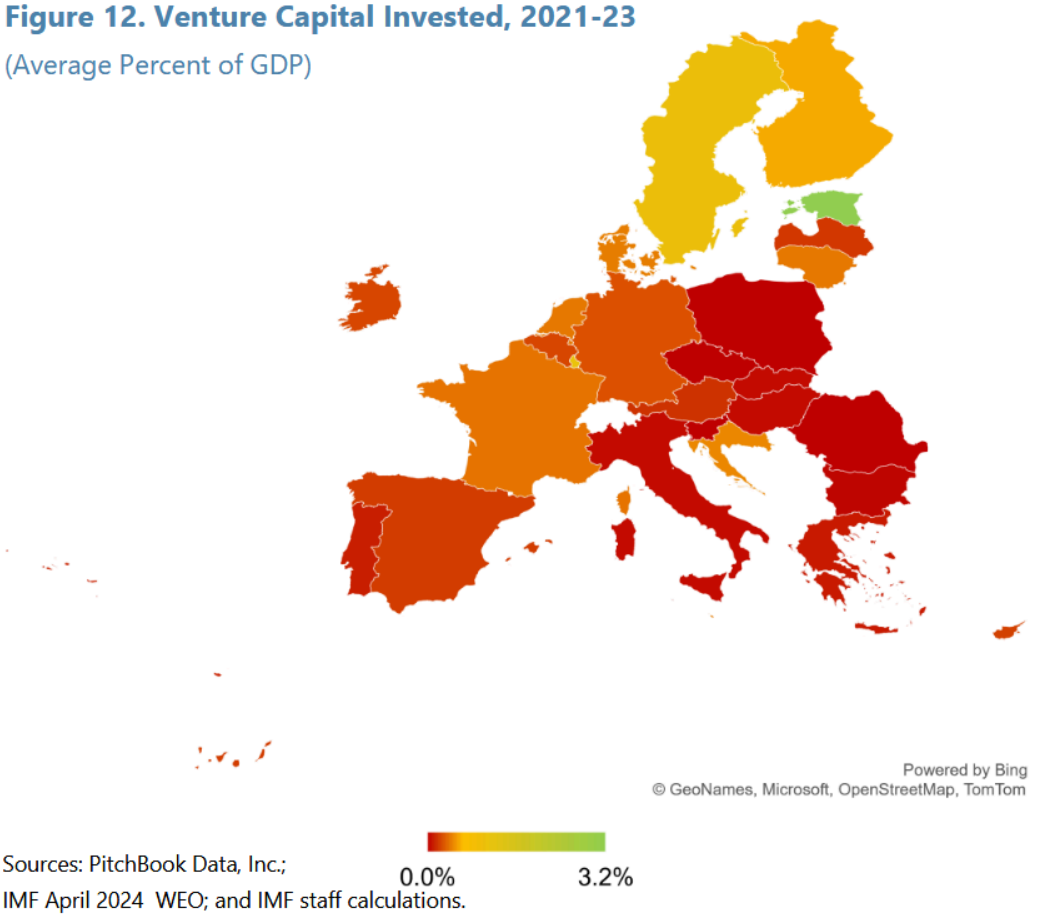


Source: ECB Flow of Funds; US Federal Reserve Flow of Funds; EBA (2015); ECB; and IMF staff calculations.
Notes: Euro area: 2022 Q4 data except for investment firms' assets which are based on EBA 2015 data, categories 1-4, 8, 10, 11. US data: 2022 Q4 data. Security brokers and dealers include holding companies, funding corporations. Mutual funds include real estate and investment trusts, exclude hedge and private equity funds.

[195] Ibid.
[196] See Footnote 19
[197] See Footnote 55
[198] See Footnote 38
[199] See Footnote 39
[200] See Footnote 67

Figure 12. Venture Capital Invested, 2021-23

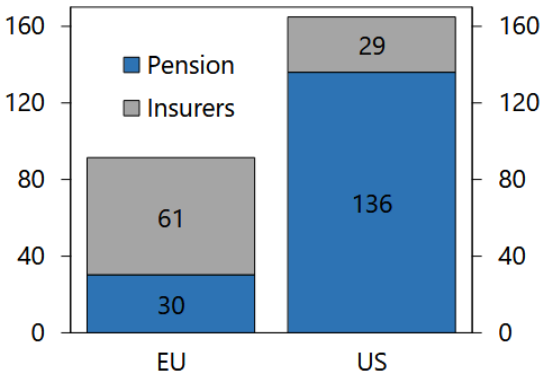
(Average Percent of GDP)



Sources: PitchBook Data, Inc.; IMF April 2024 WEO; and IMF staff calculations.

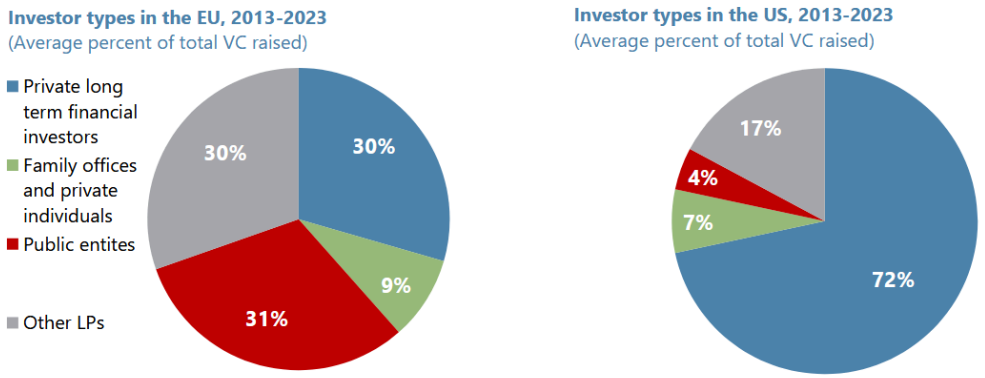
Figure 15. Pension Funds and Insurers Assets

(Percent of GDP, 2022)



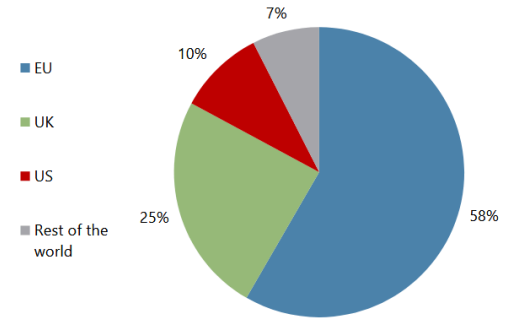
Sources: EIOPA, ECB, FRED, US Congressional Research Service; and IMF staff calculations.
Note: EU excludes AUT, BEL, BGR, CYP, HRV, MLT, ROU. Insurance data for SWE is not available.

Figure 17. Sources of VC Funds – Investor Type



Sources: PitchBook Data Inc.; IMF staff calculations

Figure 18. Sources of VC Funds – Location



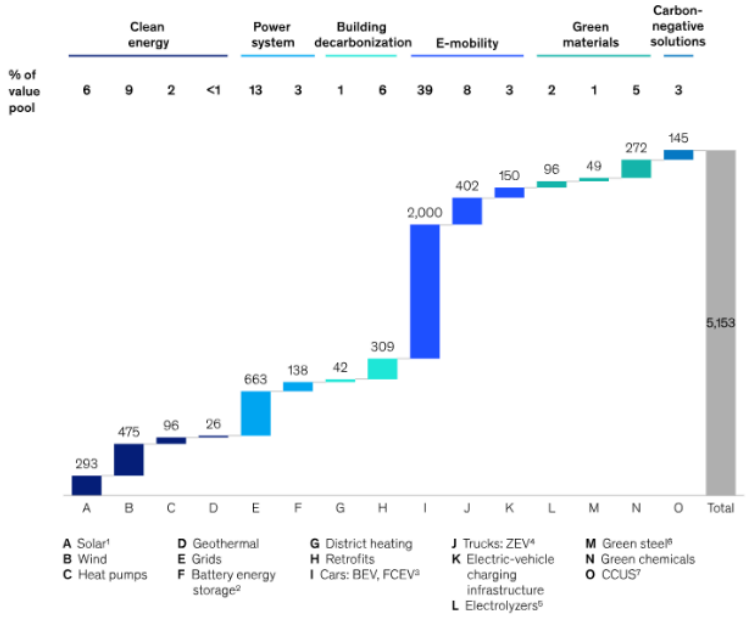
Sources: InvestEurope; and IMF staff calculations.

B - Annexes from McKinsey Sustainability Report[201]

[201] See Footnote 64

Capital expenditure spending on e-mobility makes up about half of the €5 trillion global cleantech value pool.

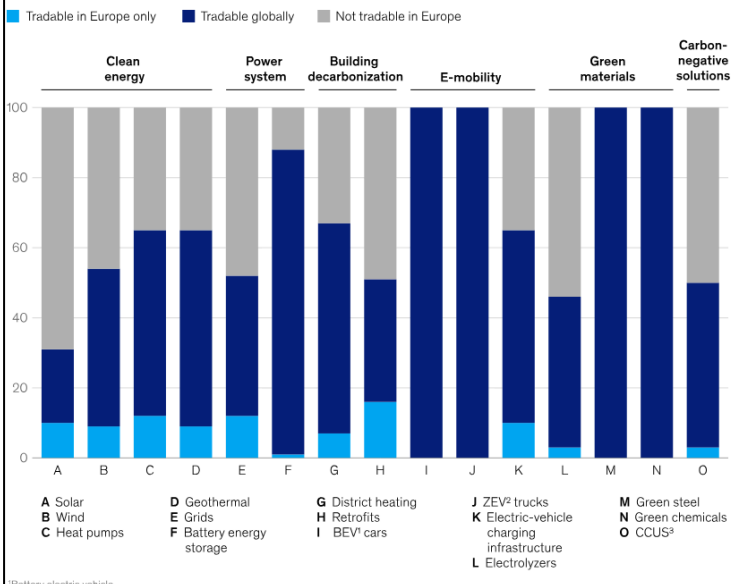
Estimated capital expenditure spending on cleantech in 2035, € billion



Note: Figures do not sum, because of rounding.
[1]Includes commercial and utility scale and distributed generation. [2]Excluding long-duration energy storage. [3]Battery electric vehicle and fuel cell electric vehicle. [4]Zero-emission vehicle. Includes BEV and FCEV. [5]Excludes other infrastructure and downstream hydrogen application. [6]Electric-arc-furnace steel. [7]Carbon capture, utilization, and storage.

McKinsey & Company

2035 annual capital expenditure spending, %



[1]Battery electric vehicle.
[2]Zero-emission vehicle.
[3]Carbon capture, utilization, and storage.

McKinsey & Company

IV - Start-ups Sectors Information

To narrow down relevant climate tech sectors, research was conducted on the most competitive and profitable sectors at the European Union level[202]. Starting with data from the Sightline 2025 report[203], the sectoral separation were as follows[204]:

1) **Energy**: the energy sector includes any technologies aimed at capturing, producing, and distributing energy. In the context of climate tech, energy has to come from renewable or non-polluting sources, such as solar, wind, geothermal, or nuclear[205].
2) **Industry**: the industrial sector comprises most manufacturing and production businesses, and often consist of the "secondary sector" of the economy[206].
3) **Built Environment**: encompassing all construction, urban planning, and architectural design necessitating labour, machinery and materials that often have non-negligeable environmental impacts[207].
4) **Transportation**: the transport sector includes any infrastructure for the movement of people or goods, as well as logistics and the construction of trade or travel routes[208].
5) **Carbon**: this sector includes any technology related to the capture, use, and storage of carbon (CCS or CCU)[209].
6) **Climate Management**: climate management includes both adaptation and mitigation to environmental changes and climate events. As this sector is relatively novel, much of the initiatives and projects within it are still quite vague and more focused on governance and establishing a coherent approach to climate management at the EU level[210].
7) **Food & Land Use**: the agricultural sector and food processing are crucial in the EU, although high competition globally has reduced its scale. Nevertheless, increasingly advanced technologies are being developed in this sector to facilitate food processing and consumption, and efficient land use[211]. Land use also includes forests and other natural habitats, which suffer from biodiversity degradation, deforestation, and land or water pollution[212].

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[202] European Commission, "CBAM Sectors," Taxation and Customs Union, 2024, https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism/cbam-sectors_en.
[203] See Footnote 68
[204] UN Environment Programme, "The Sectoral Solution to Climate Change," UN Environment Programme, 2024, https://www.unep.org/interactive/sectoral-solution-climate-change/.
[205] See Footnote 16
[206] European Commission, "Industry," Internal Market, Industry, Entrepreneurship and SMEs, 2026, https://single-market-economy.ec.europa.eu/industry_en.
[207] European Environment Agency, "Europe's Environment and Climate: Knowledge for Resilience, Prosperity and Sustainability," *European Environment Agency* (European Environment Agency, September 29, 2025), https://www.eea.europa.eu/en/europe-environment-2025/main-report/built-environment-system-with-components.
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